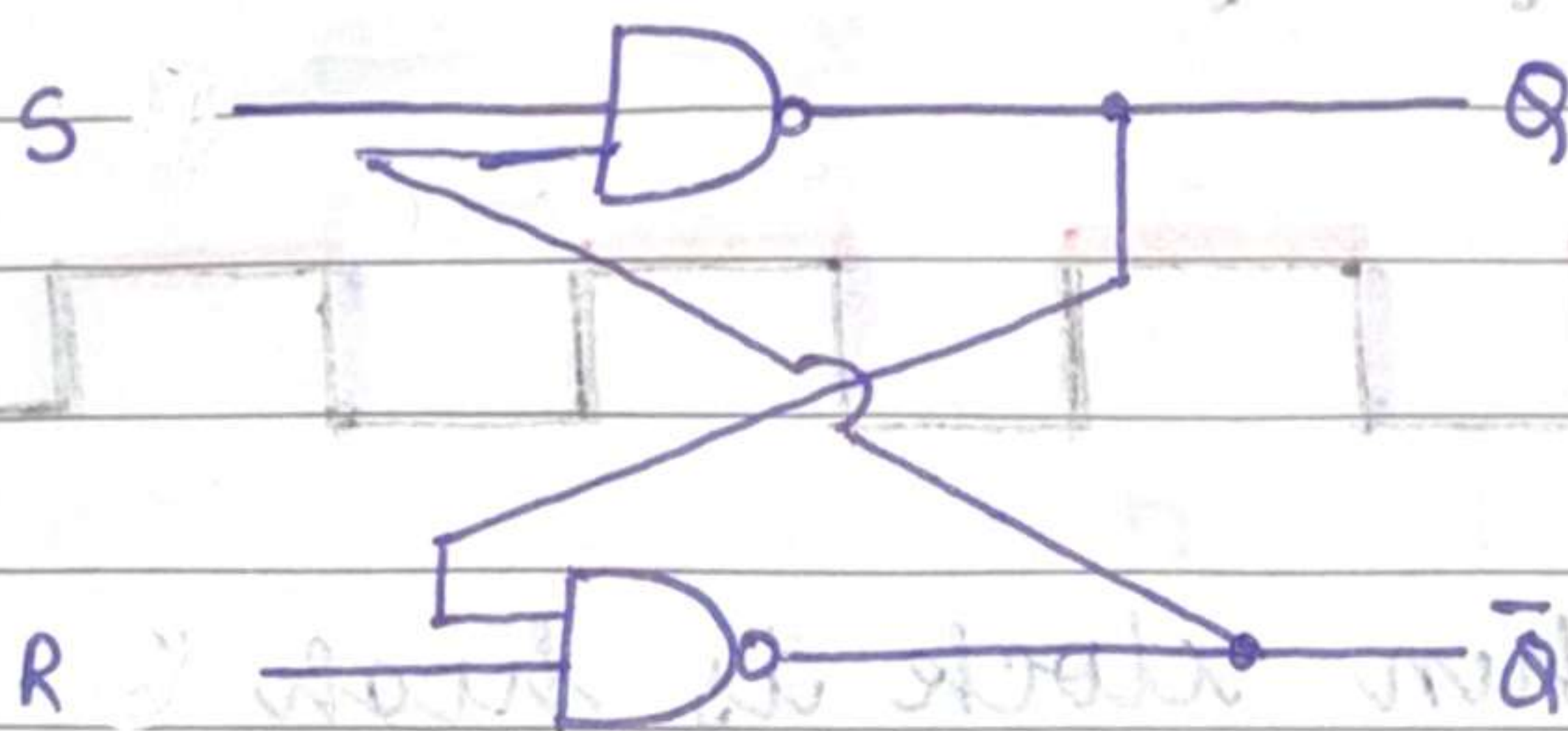


Unit → 2

→ sequential circuits

- The sequential circuits, the present output depends on the present input as well as past output or outputs
- The basic storage element is called Latch



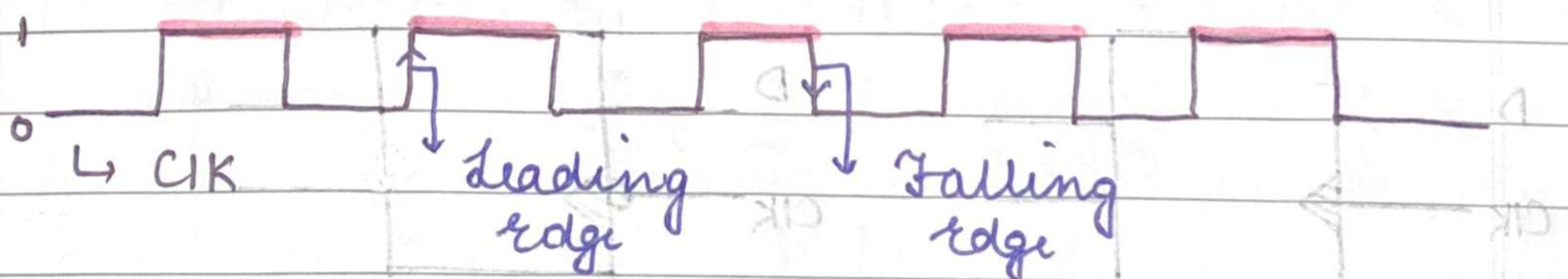
S	R	Q	$\bar{Q}$
0	0	?	?
0	1	1	0
1	0	0	1

MEMORY

→ clock

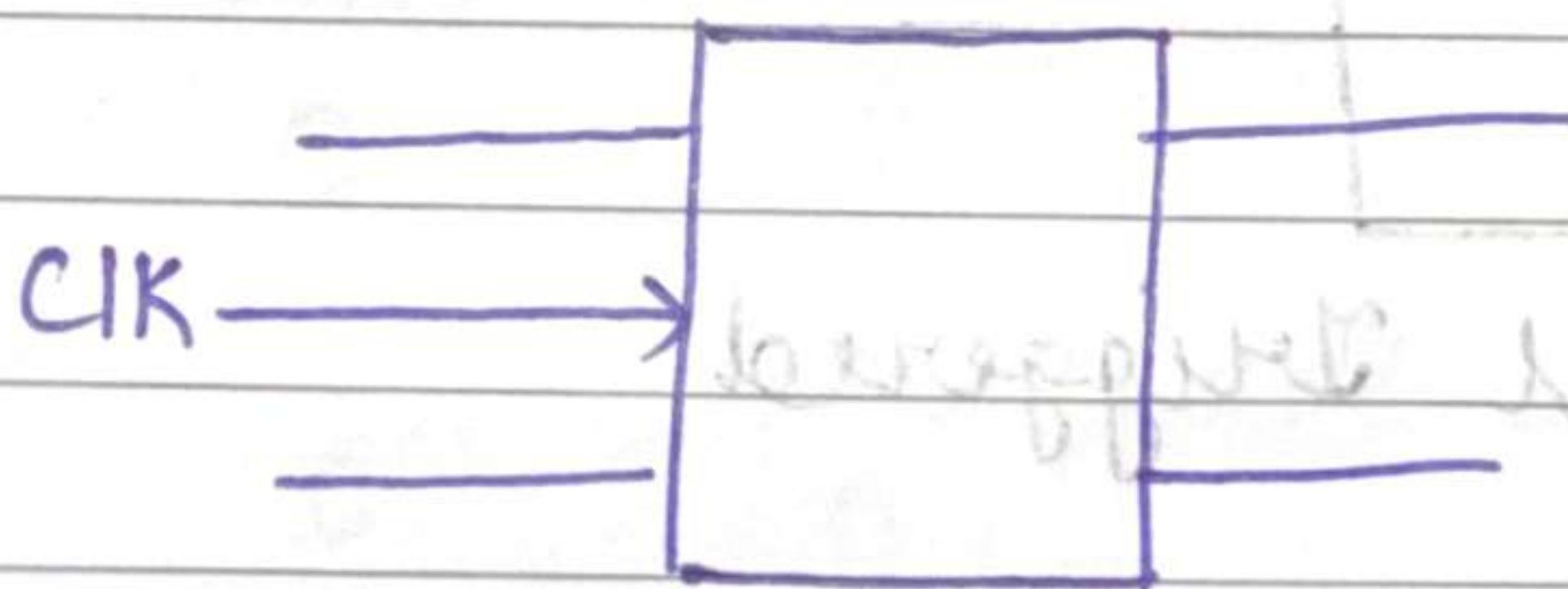
time for which signal is high
total time

- A signal (having duty) cycle of 50% and controls the working of circuit



Flip flop functional only when clk is high (—)

or other cases also (edge triggering)
i.e. high to low
or low to high } transitions



→ Triggering methods

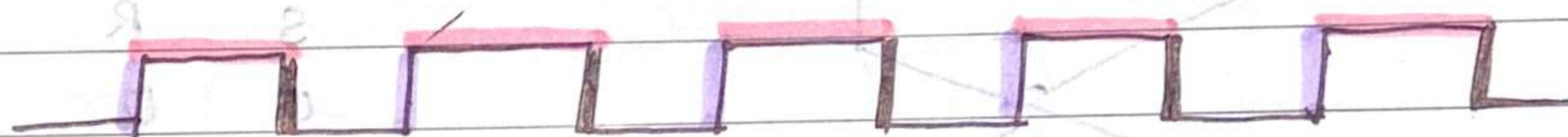
- The mechanism by which the flip flops change their memory state according to clock pulse and input is called as triggering

1. Level triggering (—)

2. edge triggering (—)

↳ +ve edge triggering (—)

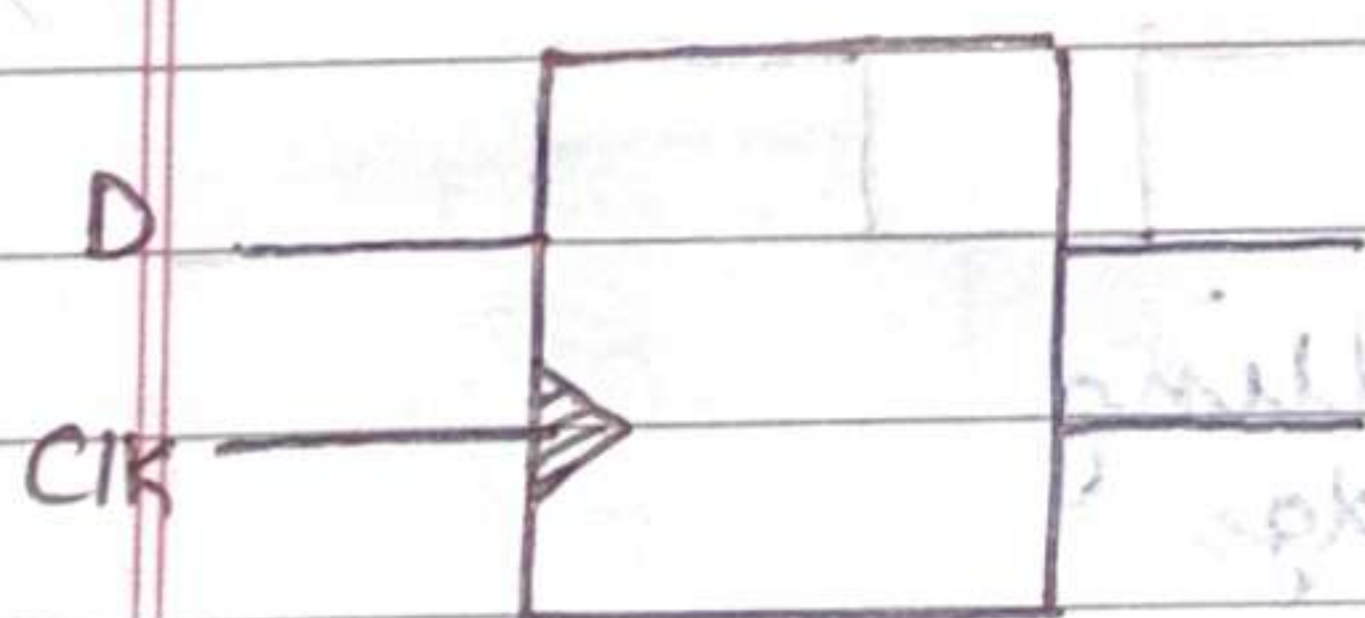
↳ -ve edge triggering (—)



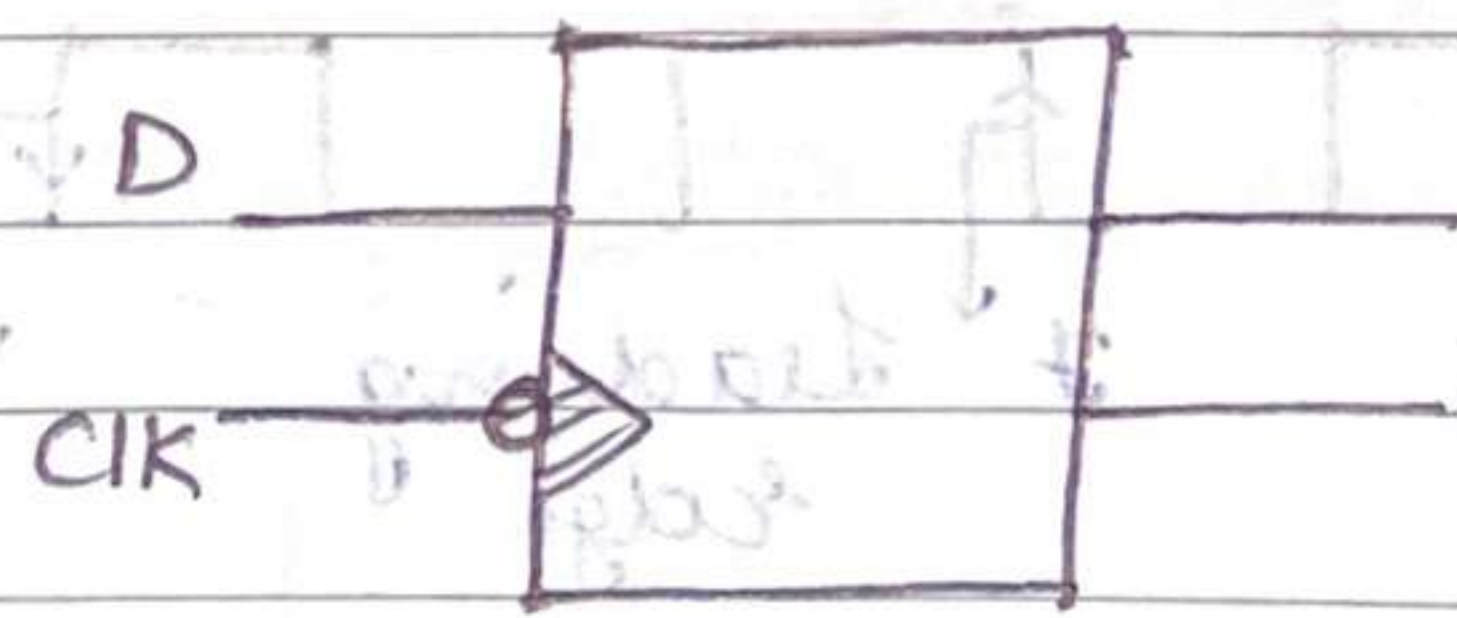
• Level → only when clock is high

• +ve edge → while transitioning from low to high (level up)

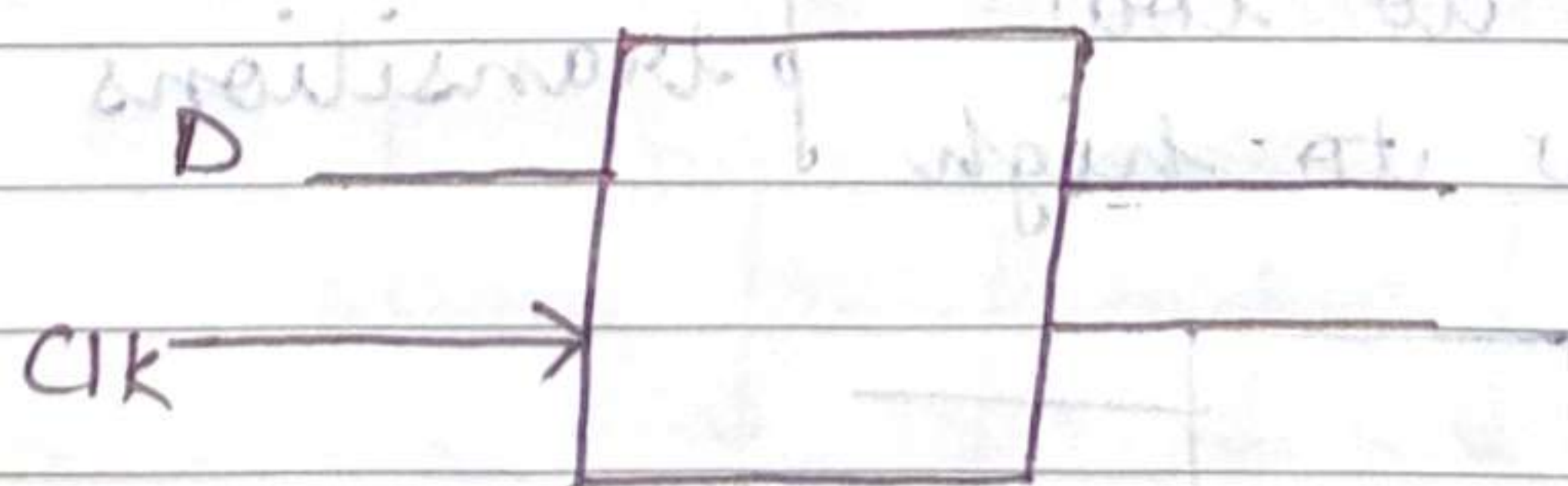
• -ve edge → while transitioning from high to low (falling down)



+ve edge
Triggered

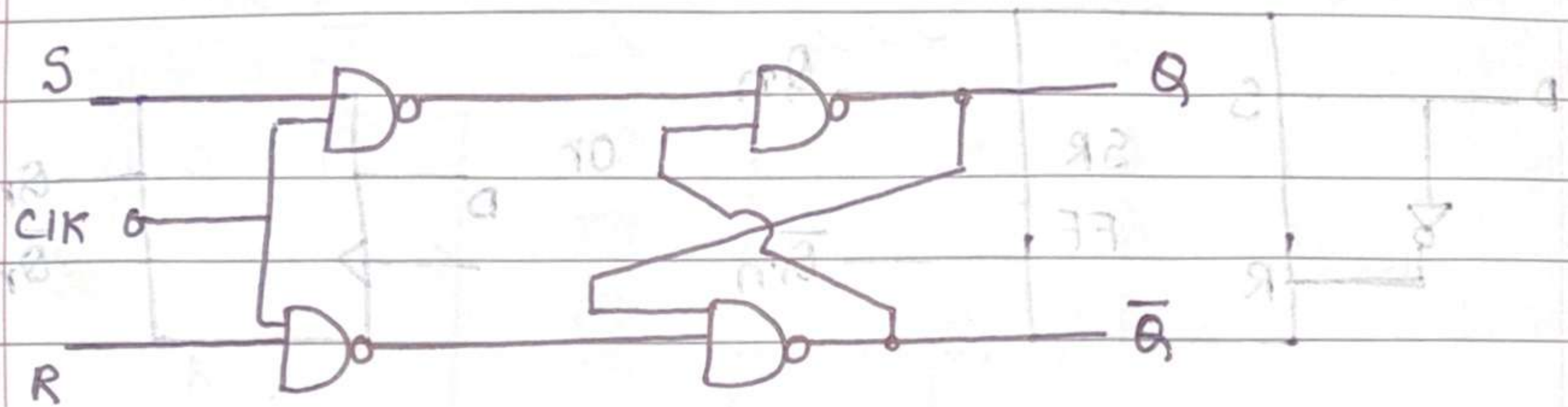


-ve edge
triggered



Level Triggered

→ SR Flip Flop



Clock	S	R	Q_{n+1}
0	X	X	Q_n
1	0	0	Q_n
1	0	1	0
1	1	0	1
1	1	1	?

→ Characteristic Table for SR flip flop

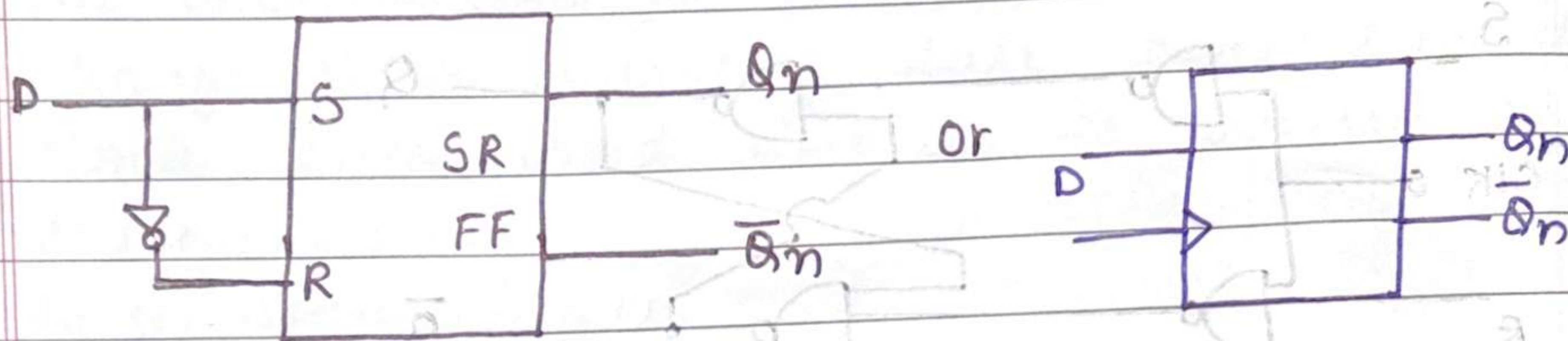
- Used to find next state using present state

Q_n	S	R	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	X
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	X

→ Excitation Table for SR flip flop

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

↳ uses outputs to find inputs
Helpful in flip flop conversions and counters

→ SR Flip Flop

Clock Q_{n+1}

0

X

 Q_n

1

0

0X

1

1

10

→ Characteristic Table for SR flip flop

Q_n D Q_{n+1}

0

0

0

0

1

1

 $Q_{n+1} = D$

1

0

0

1

1

1

→ Excitation Table for SR flip flop Q_n Q_{n+1}

X

D

0

0

1

0

0

1

0

1

1

0

1

0

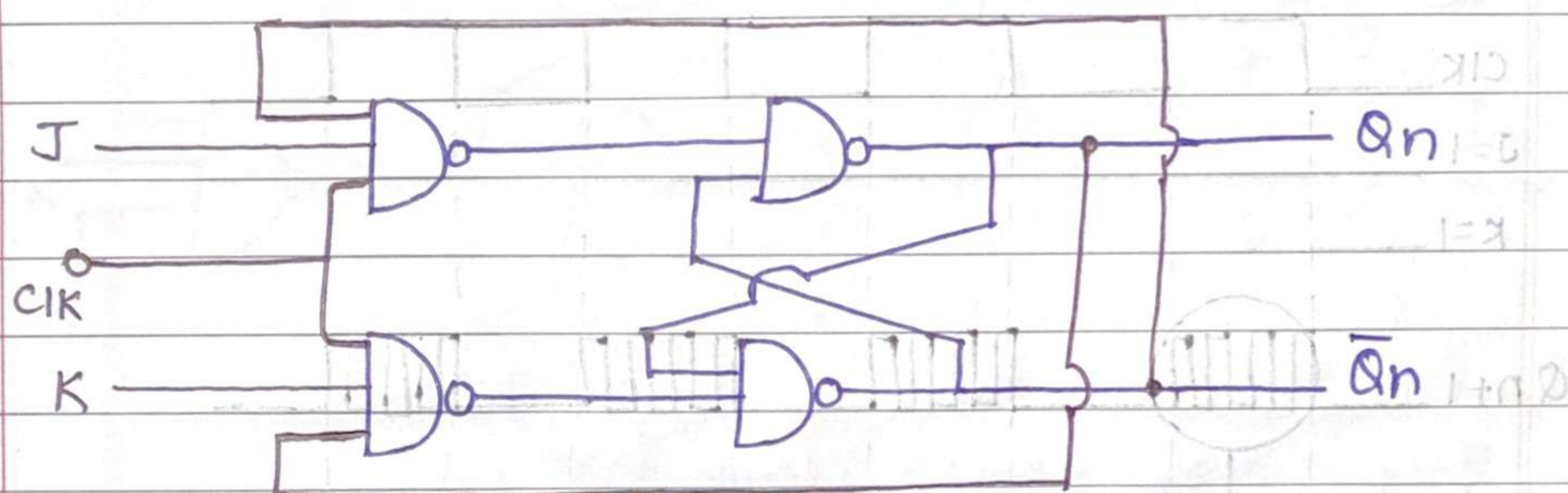
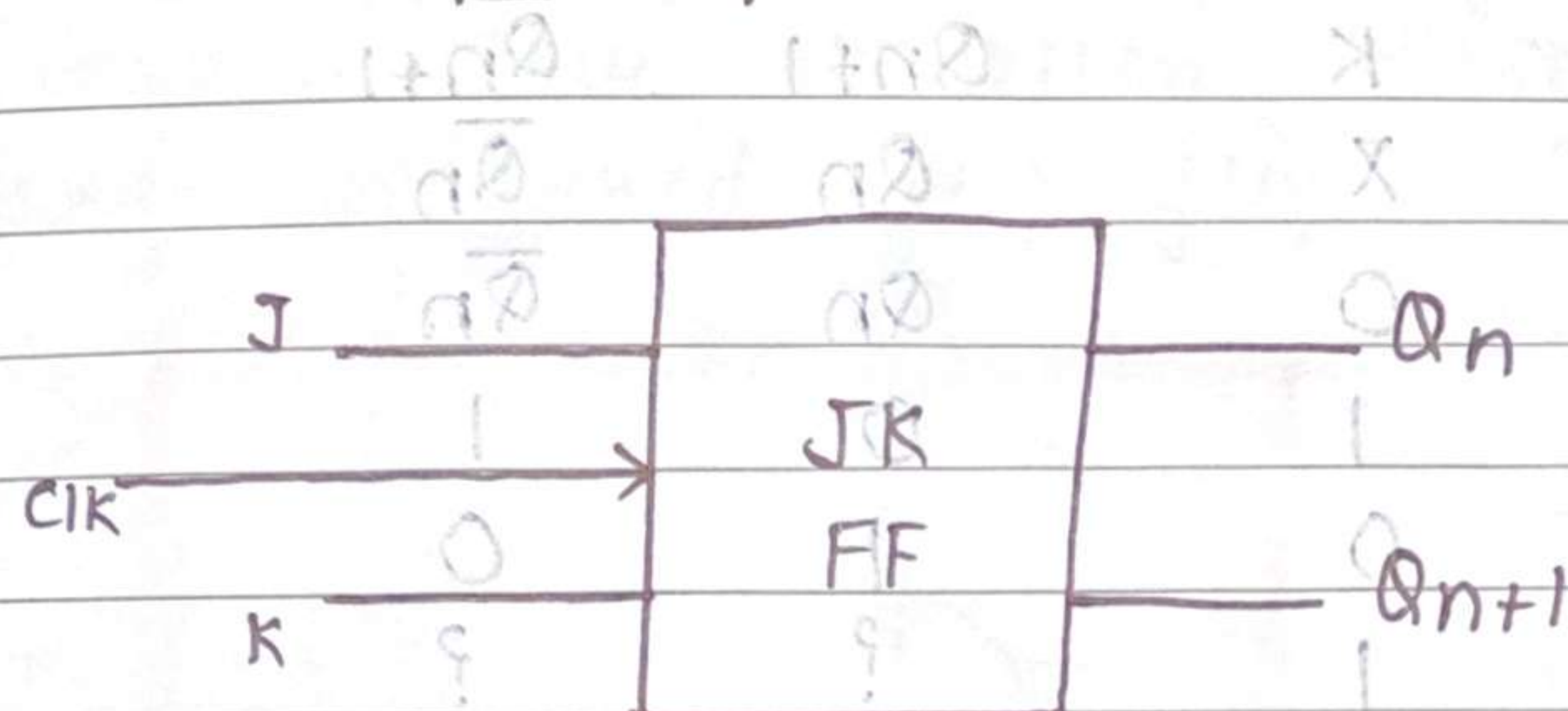
1

1

X

1

→ JK Flip Flops



Clock	J	K	Q_{n+1}
0	X	X	Q_n
1	0	0	Q_n
1	0	1	0
1	1	0	1
1	1	1	$\bar{Q}_n$

→ Characteristic Table

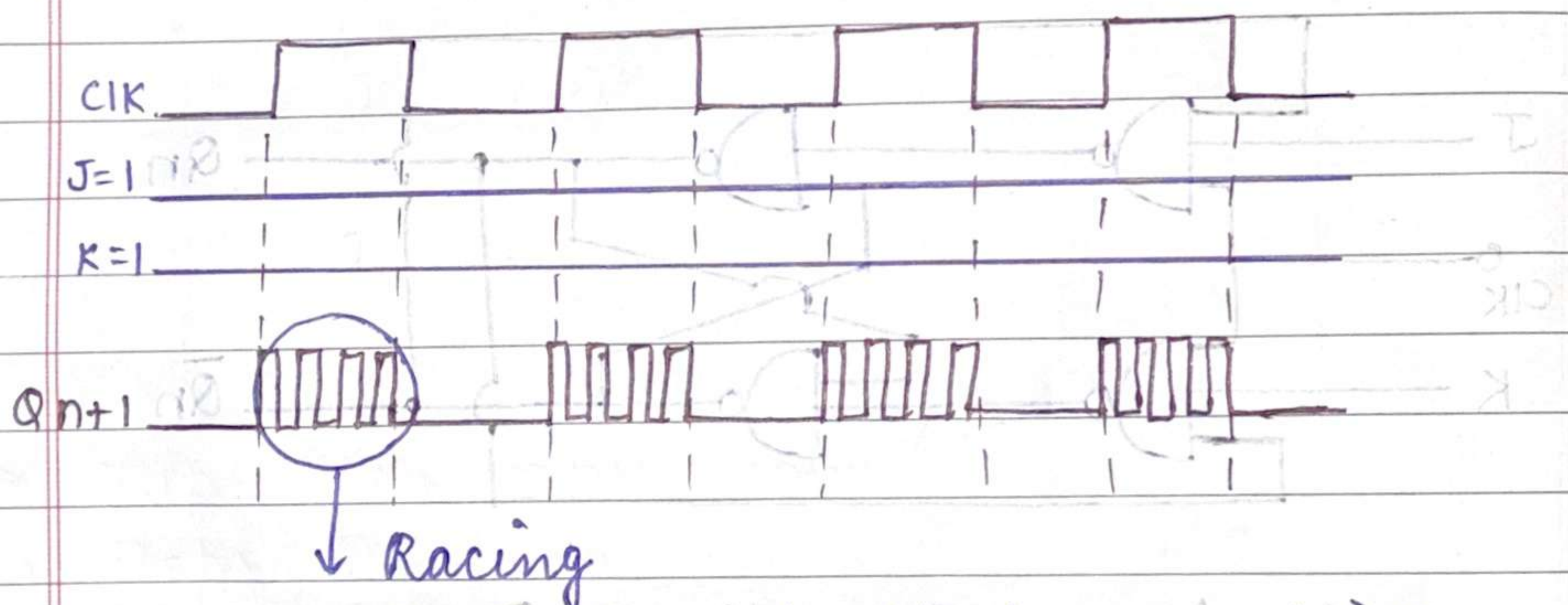
Q_n	J	K	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

Excitation Table

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

→ Race around conditions

clock	J	K	Q_{n+1}	$\bar{Q}_{n+1}$
0	X	X	Q_n	$\bar{Q}_n$
1	0	0	Q_n	$\bar{Q}_n$
1	0	1	0	1
1	1	0	1	0
1	1	1	?	?



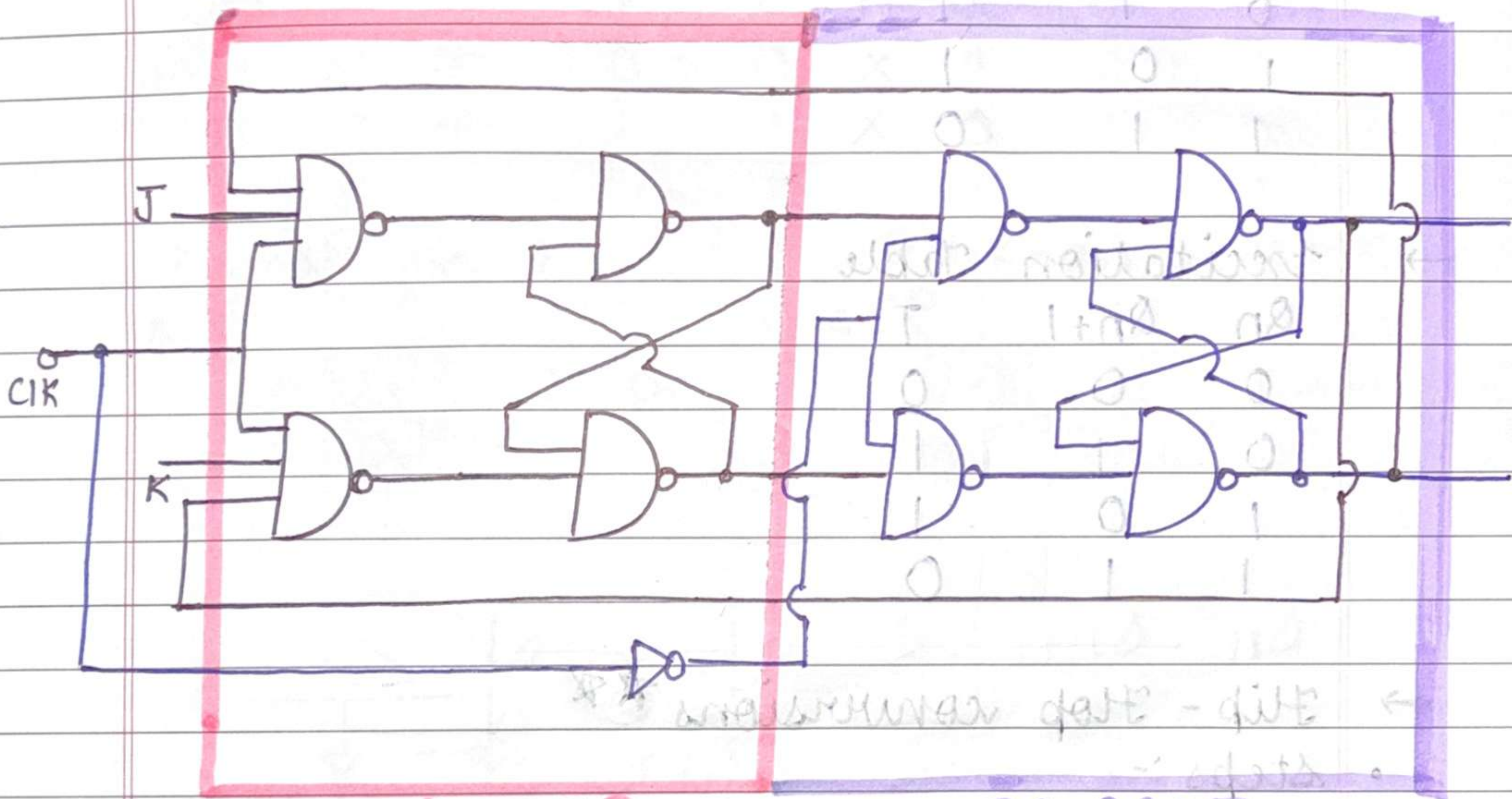
When all J, K and clock are high, the output state switches between 0 & 1 multiple times until the clock becomes low. This condition is called as race around condition.

→ Conditions to overcome racing

- $T/2 < \text{propagation delay of FF}$ ($T \rightarrow \text{clock frequency}$)
- Using edge triggering
- Master slave J-K flip flop

X	1	1	0	0	1	0	0
1	X	0	1	1	1	1	0
0	X	1	1	1	0	0	1
				0	1	0	1
				1	0	1	1
				0	1	1	1

- Master Slave J-K flip flop
- Master slave operation is similar to negatively edge triggered flip flop

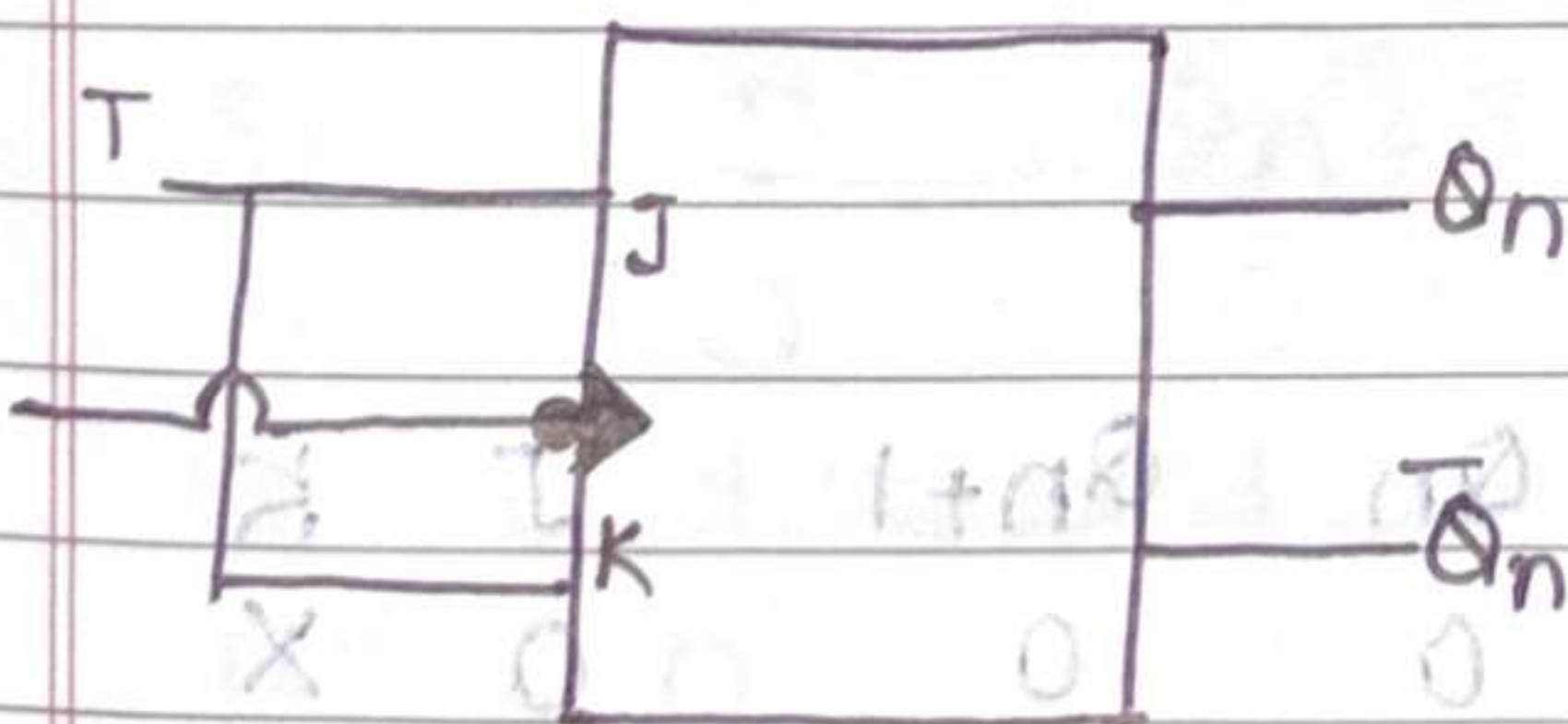
**MASTER****SLAVE**

→ Active when
clk is high

→ Active when
clk is low

It is used to remove the race-around cond by using toggling

→ T Flip Flop



CLK	T	Q_{n+1}
0	X	Q_n
1	0	Q_n
1	1	$\bar{Q}_n$

→ Characteristic Table

Q_n	T	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

→ Excitation Table

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

→ Flip-Flop conversions ★★

• Steps :-

1. Identify available & required flip flop
2. Make characteristic table for required flip flop
3. Make excitation Table for available flip flop
4. Write boolean expression of available flip flop
5. Draw the circuit.

1. JK to D → required

Q_n	D	Q_{n+1}
0	0	0
0	1	1
1	0	0
1	1	1

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Now

Q_n	D	Q_{n+1}	J	K
0	0	0	0	X
0	1	1	0	1
1	0	0	X	1
1	1	1	X	0

K map for J

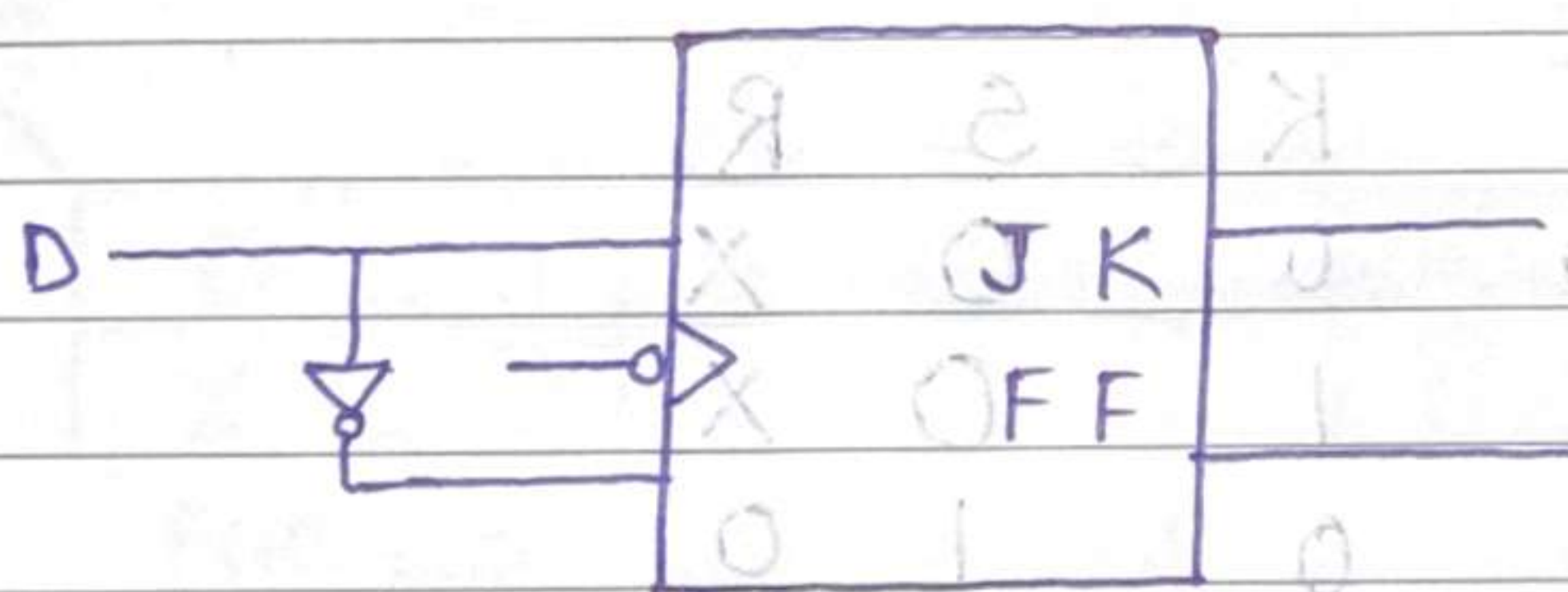
$Q_n \backslash D$	0	1
0	0	1
1	X	X

$$J = D$$

K-map for K

$Q_n \backslash D$	0	1
0	X	X
1	1	0

$$K = \bar{D}$$



2. T flip flop to D flip flop

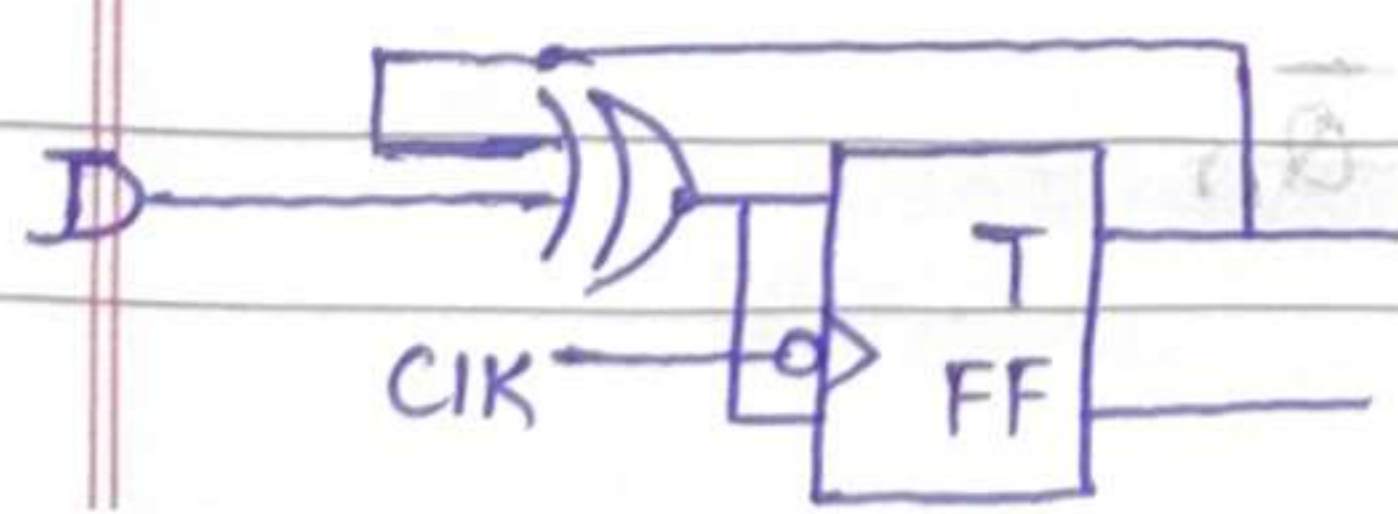
Q_n	D	Q_{n+1}	T
0	0	0	0
0	1	1	0
1	0	0	1
1	1	1	0

$Q_n \backslash D$	0	1
0	0	0
1	0	1

$Q_n \backslash D$	0	1
0	0	1
1	1	0

$$T = \bar{D}Q_n + Q_n D$$

$$\Rightarrow T = D \oplus Q_n$$



→ SR to JK Flip Flop

Q_n	J	K	Q_{n+1}	S	R
0	0	0	0	0	X
0	0	1	0	1	0
0	1	0	1	0	1
0	1	1	1	X	0
1	0	0	1	0	0
1	0	1	0	0	1
1	1	0	1	0	0
1	1	1	0	1	0

Q_n	Q_{n+1}	J	K	S	R
0	0	0	0	0	X
0	0	0	1	0	X
0	1	1	0	1	0
0	1	1	1	1	0
1	1	0	0	X	0
1	0	0	1	0	1
1	0	1	0	X	0
1	1	1	1	0	1

Q_n	$\bar{J}\bar{K}$	$\bar{J}K$	$J\bar{K}$	JK
$\bar{Q}_n$	0	0	1	1
Q_n	X	0	0	X

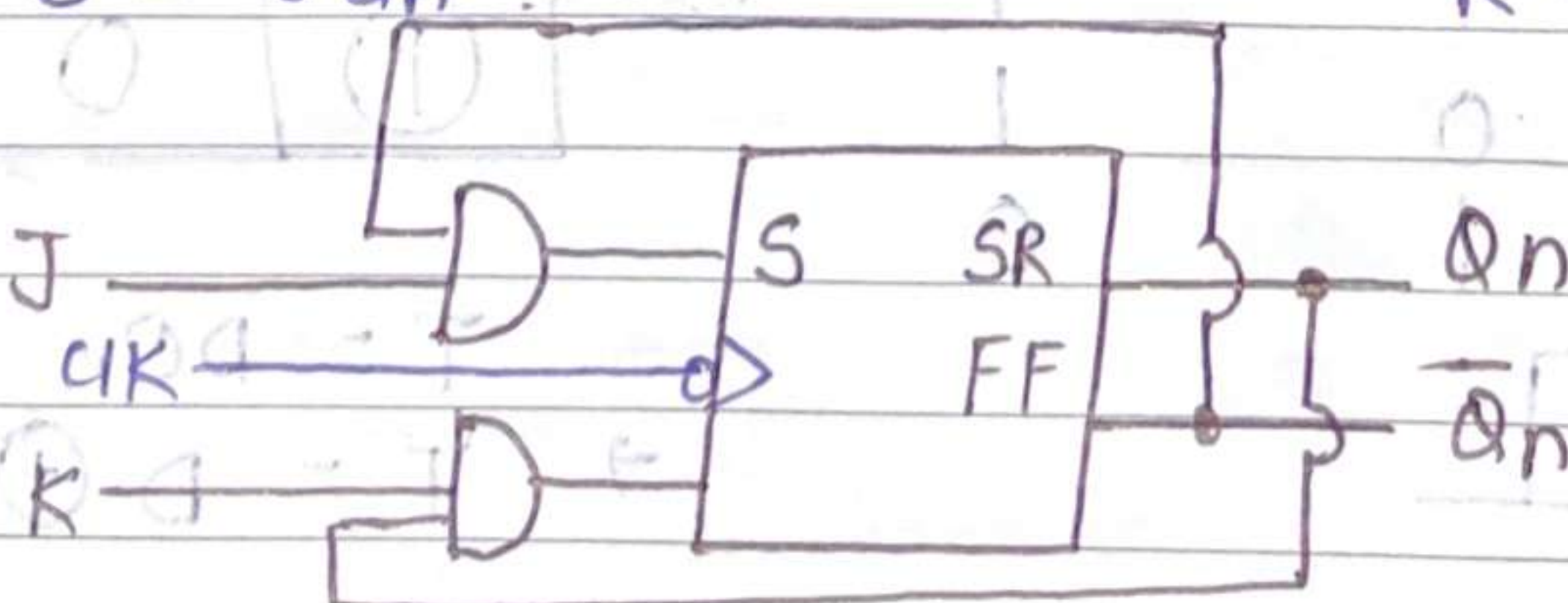
For S

$$S = J\bar{Q}_n$$

Q_n	$\bar{J}\bar{K}$	$\bar{J}K$	$J\bar{K}$	JK
$\bar{Q}_n$	X	X	0	0
Q_n	0	1	1	0

For R

$$R = Q_n K$$



→ SR to T flip flop

Q_n	T	Q_{n+1}	Q_n	Q_{n+1}	S	R
0	0	0	0	0	0	X
0	1	1	0	1	1	0
1	0	0	1	0	0	1
1	1	0	1	0	X	0

Q_n	T	Q_{n+1}	S	R
0	0	0	0	X
0	1	1	1	0
1	0	0	X	0
1	1	0	0	1

Q_n	T
0	X
X	0

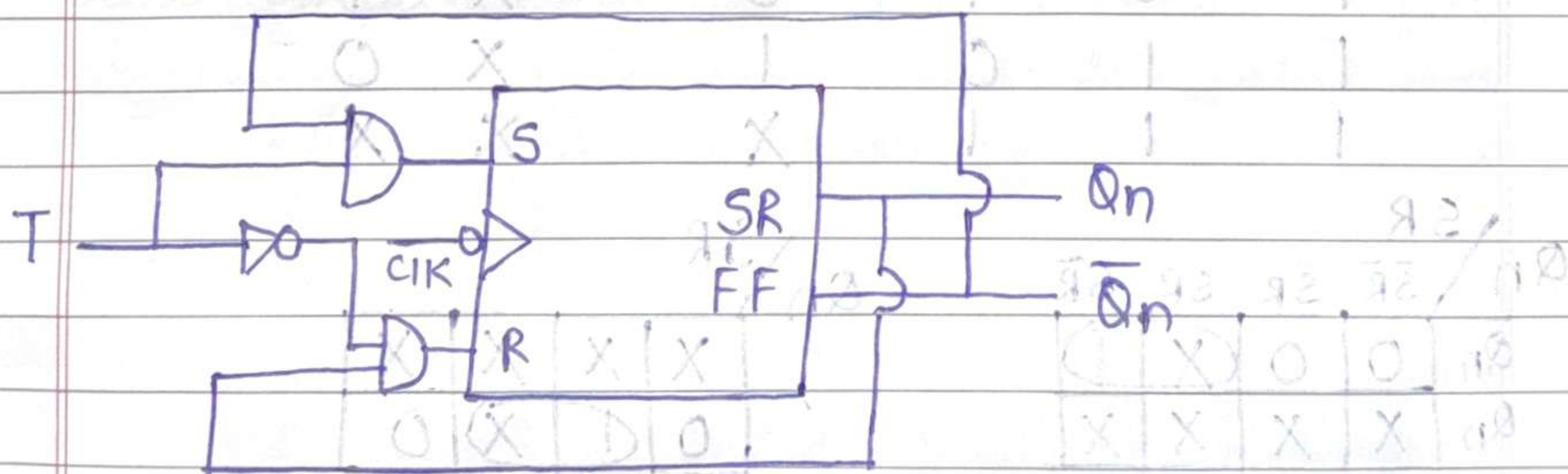
For S

$$S = T \bar{Q}_n$$

Q_n	T
X	0
0	0

For R

$$R = \bar{T} Q_n$$



→ JK to SR FF conv. diff. T table

Q_n	S	R	Q_{n+1}	Q_n	Q_{n+1}	J	K
0	0	0	0	0	0	0	X
0	0	1	0	0	0	1	X
0	1	0	1	0	1	X	1
0	1	1	X	1	1	X	0
1	0	0	1	1	1	0	X
1	0	1	0	1	0	0	0
1	1	0	1	1	1	0	0
1	1	1	X	1	0	1	1

Q_n	S	R	Q_{n+1}	J	K
0	0	0	0	0	X
0	0	1	0	0	X
0	1	0	1	1	X
0	1	1	X	X	X
1	0	0	1	X	0
1	0	1	0	X	1
1	1	0	1	X	0
1	1	1	X	X	X

Q_n	$\bar{S}R$	$\bar{S}\bar{R}$	SR	$S\bar{R}$
0	0	0	X	1
1	X	X	X	X

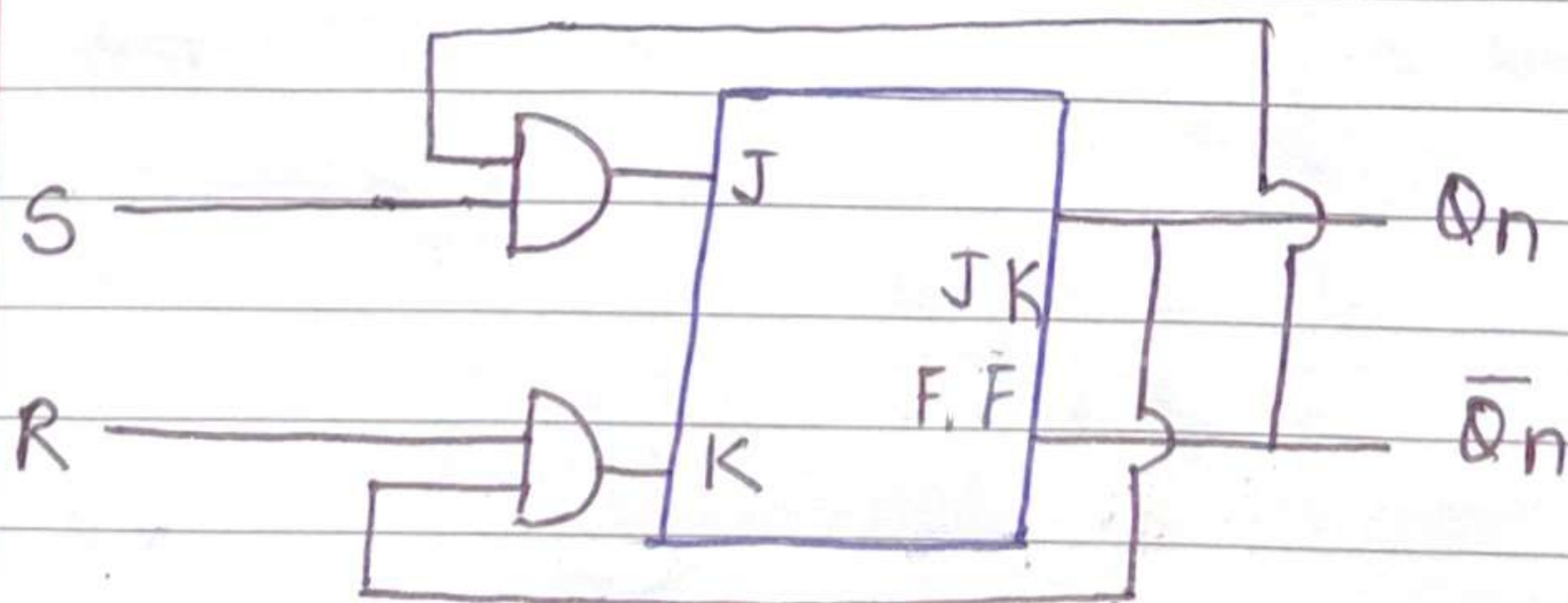
For J

$$J = S\bar{Q}_n$$

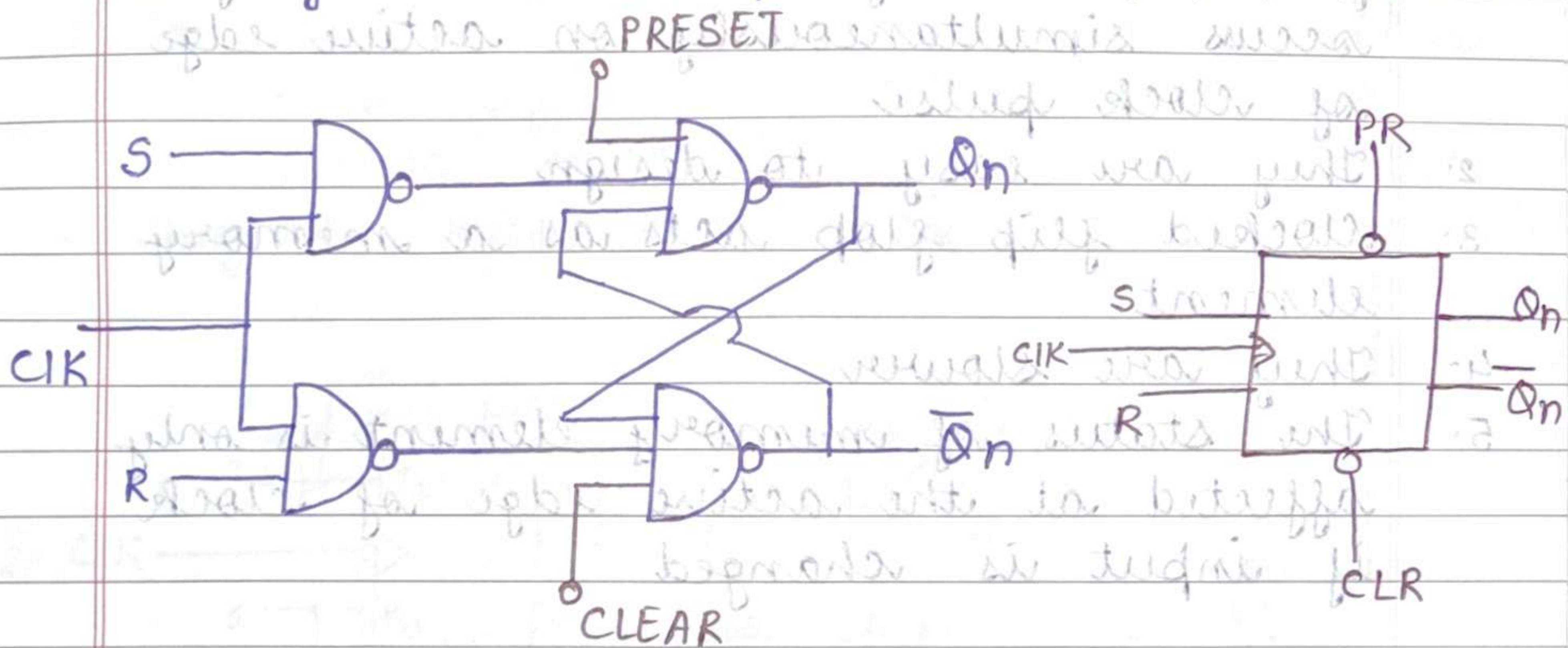
Q_n	$\bar{S}R$	$\bar{S}\bar{R}$	SR	$S\bar{R}$
X	X	X	X	X
0	1	X	0	0

For K

$$K = RQ_n$$



- Preset and Clear Inputs
- They are the direct inputs / overriding inputs or asynchronous inputs
 - The synchronous inputs are S, R, J, K, D & T



When Preset = 0, $Q_n = 1$ always
 When Clear = 0, $Q_n = 0$ always
 irrespective of value of clk, S, R, J, K, D, T
 PR & CLR are active low inputs

PR	CLR	Q_n
0	0	Not used
0	1	1
1	0	0

1 threshold FF performs normally

- Applications of flip flops
1. Registers & shift registers
 2. Binary counters
 3. Frequency division
 4. Memory units
 5. Data Synchronisation
 6. Switch debouncing

→ Synchronous sequential circuits

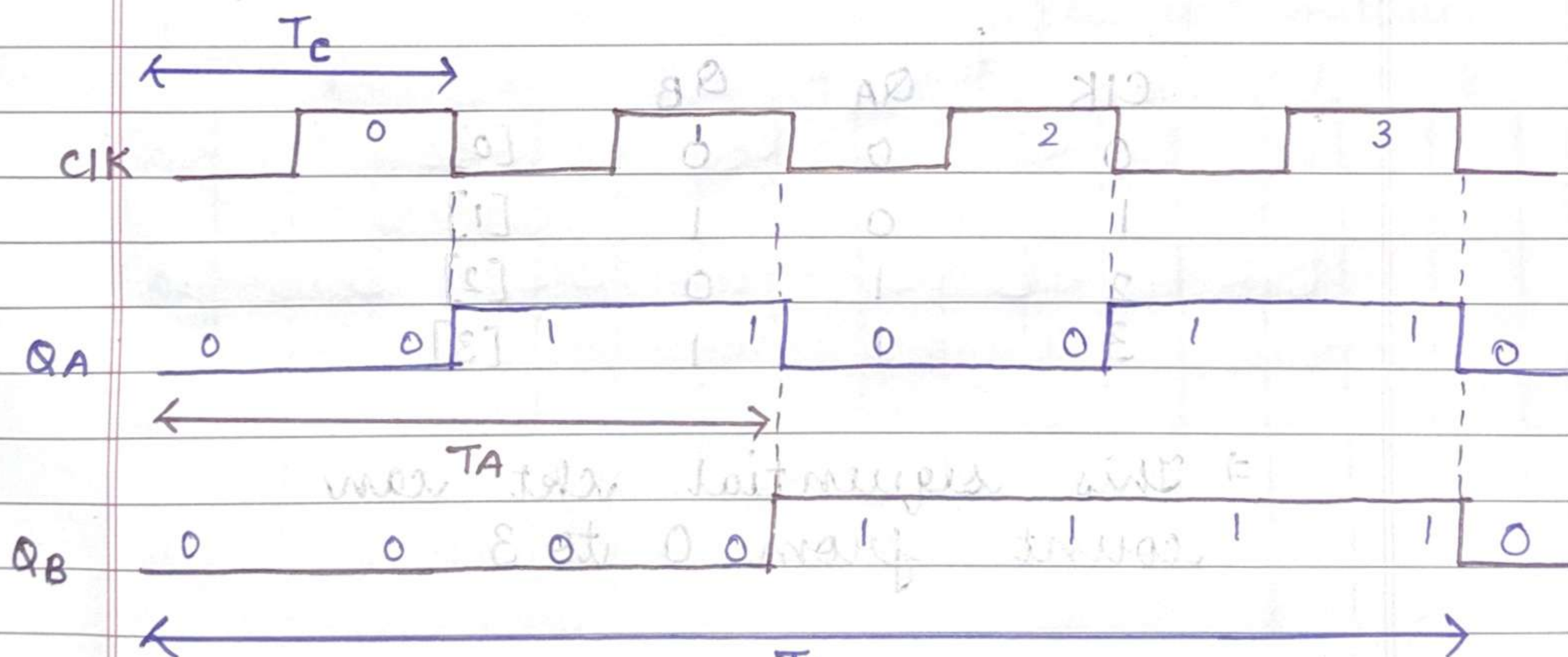
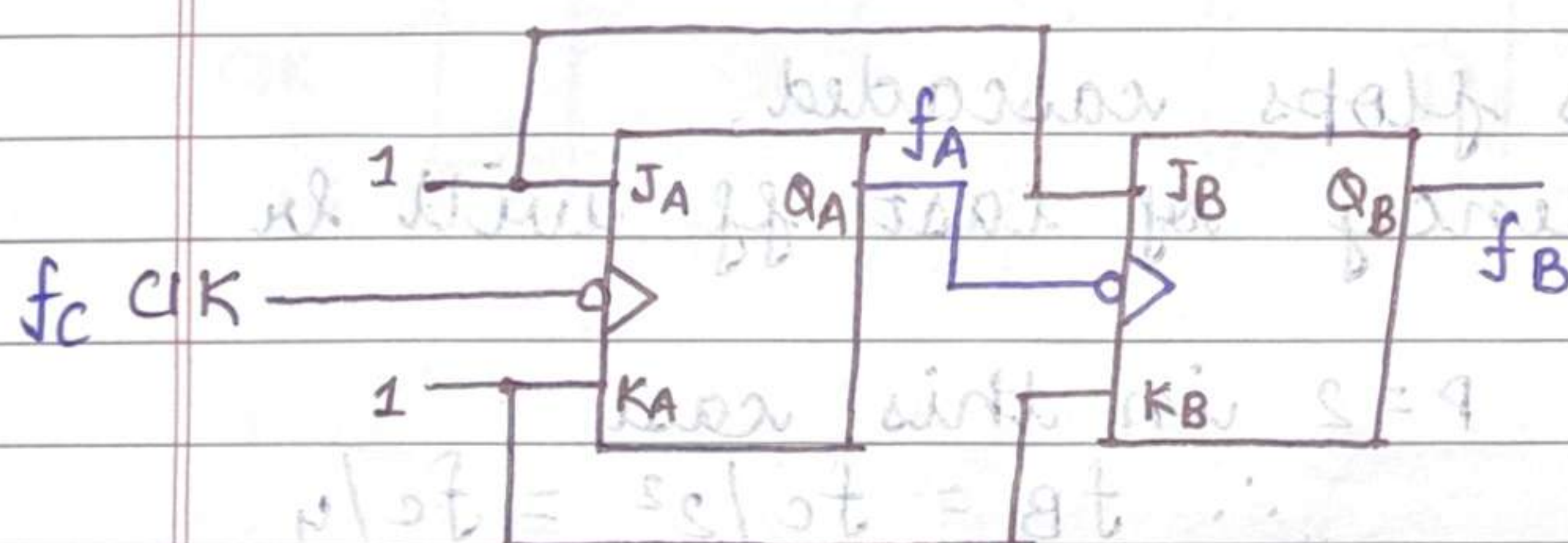
1. They are the circuits in which all memory elements (ff) share the single common global clock signal & state change occurs simultaneously on active edge of clock pulse
2. They are easy to design
3. Clocked flip flop acts as a memory element
4. They are slower
5. The status of memory element is only affected at the active edge of clock if input is changed

→ Asynchronous sequential circuits

1. They have no global clock signal instead the memory elements change their state immediately in response to external input transitions or when output of one stage triggers the input of next
2. They are difficult to design
3. An unclocked flip flop / time delay element is used as a memory element
4. Faster as clock is absent
5. The status of memory element will change anytime as soon as the input is changed

COUNTERS

- Counters are devices that can be used to count a sequence on nos like 1-10 or 2-6 as per our needs
- They use flip flops as divide by 2 circuits



- As we used -ve edge triggering state only changes in fall down
- Q_A acts as clock for Q_B

By timing diagram, $T_A = 2T_c$

$$\Rightarrow \frac{1}{f_A} = \frac{2}{f_c} \Rightarrow f_c = 2f_A$$

$$\Rightarrow f_A = \frac{f_c}{2}$$

Also $t_B = 2t_A$

$$\Rightarrow \frac{1}{f_B} = \frac{2}{f_A}$$

$$\Rightarrow f_B = \frac{f_A}{2}$$

$$\Rightarrow f_B = \frac{f_c}{4}$$

For p flip flops cascaded,
the frequency of last ff will be
 $\frac{f_c}{2^p}$

$p = 2$ in this case

$$\therefore f_B = f_c / 2^2 = f_c / 4$$

CLK	Q_A	Q_B	
0	0	0	[0]
1	0	1	[1]
2	1	0	[2]
3	1	1	[3]

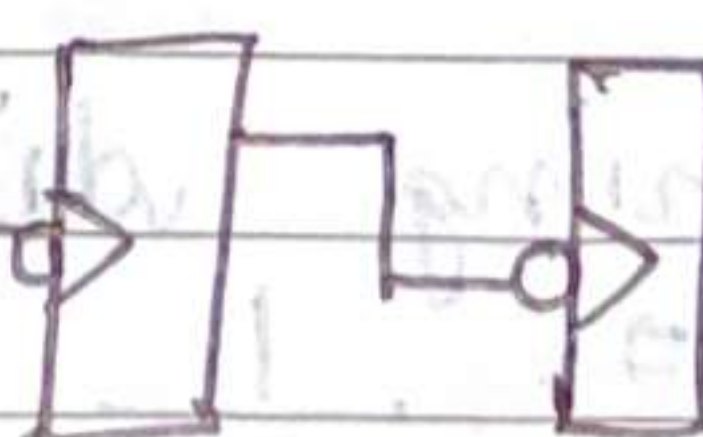
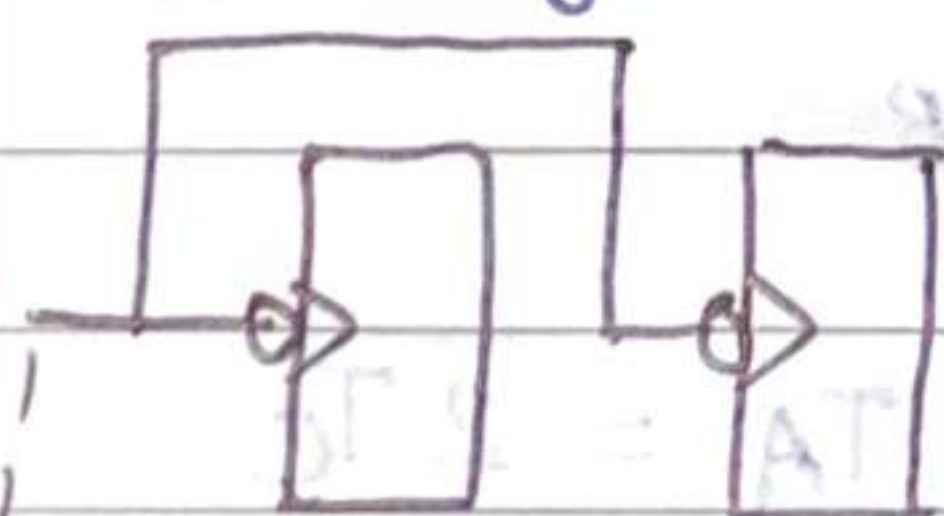
$\Rightarrow$ This sequential ckt can
count from 0 to 3

$\rightarrow$ Types of counters

Synchronous

Asynchronous

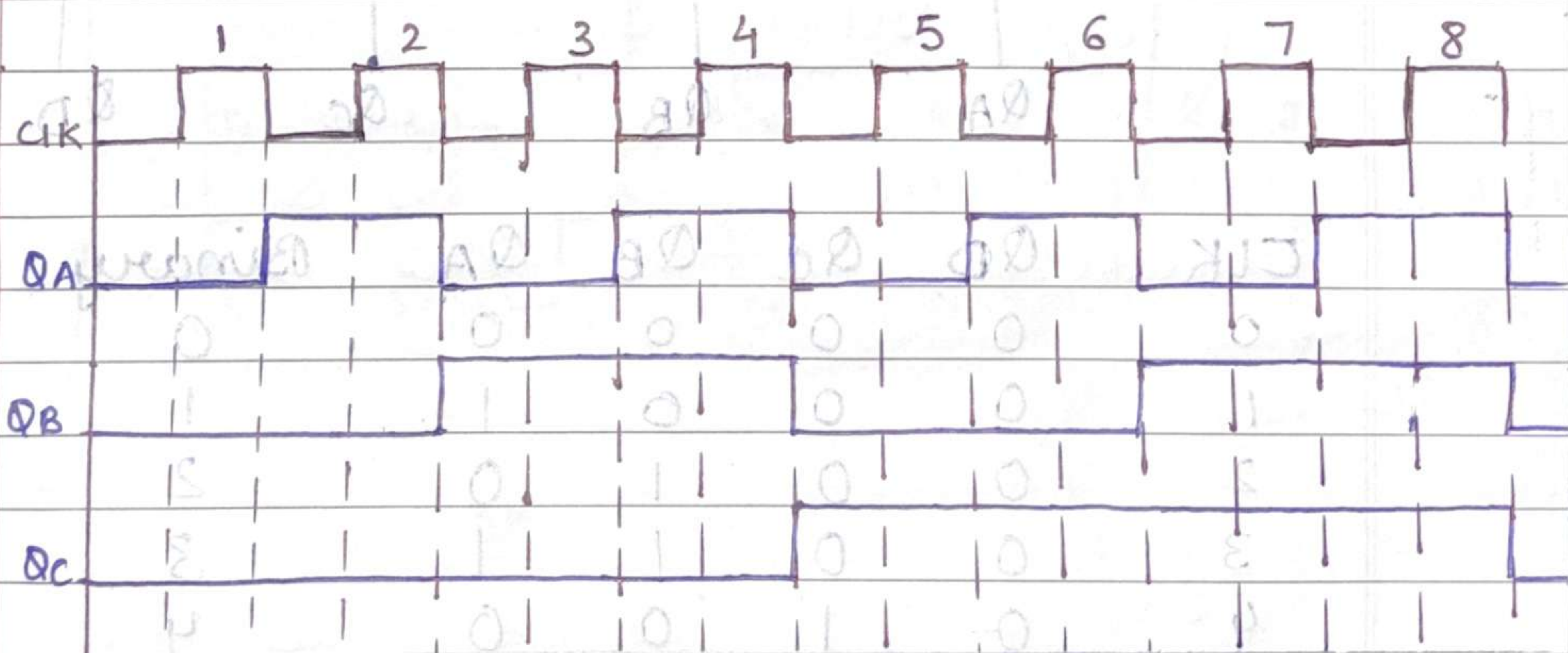
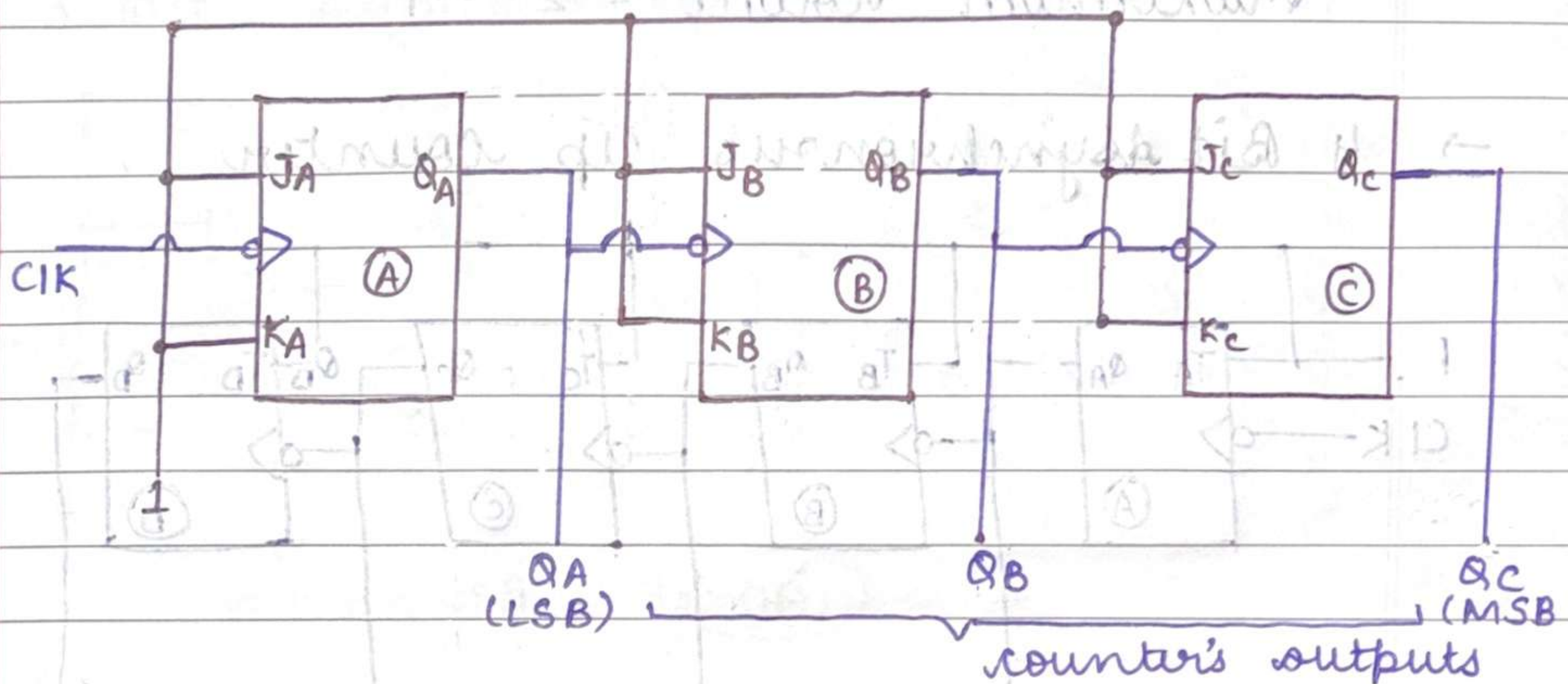
(Ripple counter)



Both syn. & asyn. counters can be :-

1. up (0, 1, 2, 3)
2. down (3, 2, 1, 0)
3. up/down

→ 3 bit Asynchronous up counter

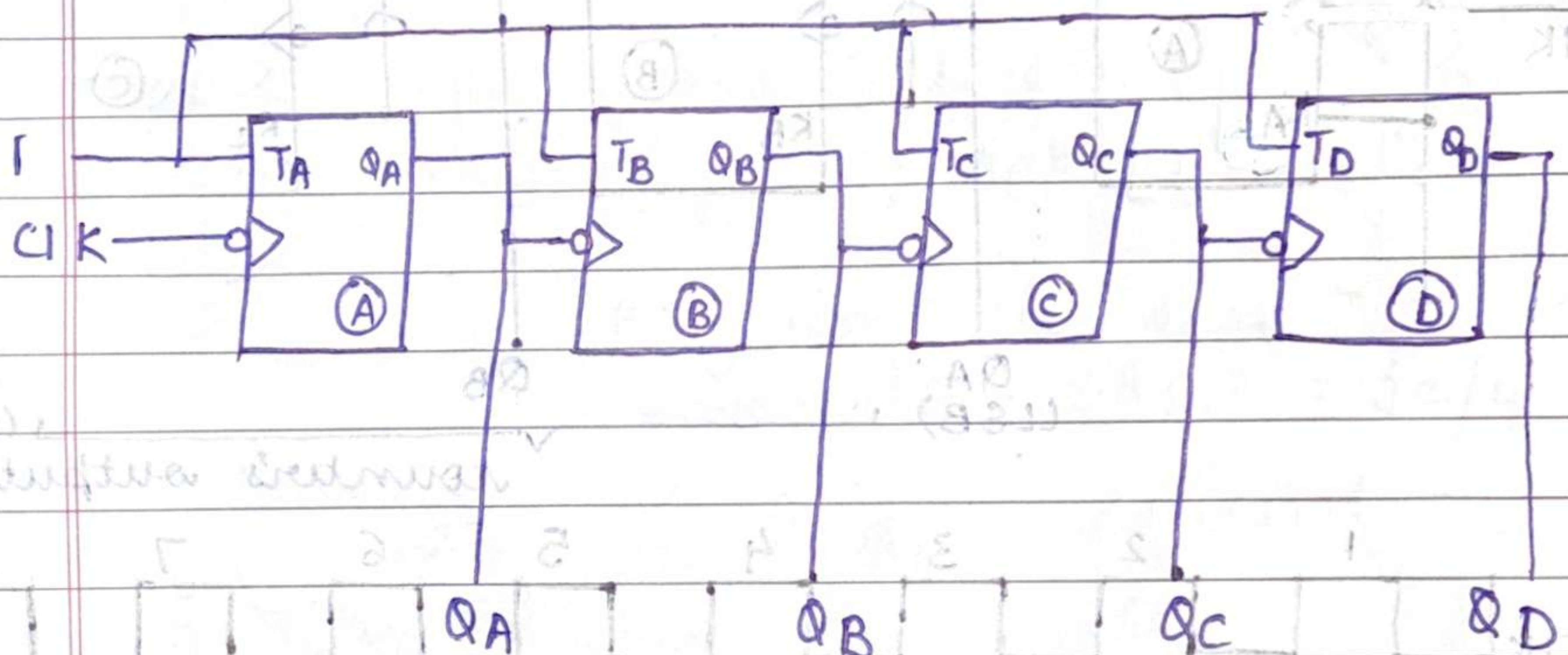


CLK	QC	QB	QA	decimal
(initially) 0	0	0	0	0
1	0	0	1	1
2	0	1	0	2
3	0	1	1	3
4	1	0	0	4
5	1	0	1	5

CLK	Q _C	Q _B	Q _A	Decimal
6	1	1	0	(1, 1, 0)
7	1	1	1	(0, 1, 1)
8	0	0	0	0

For n states,
Maximum count = $2^n - 1$

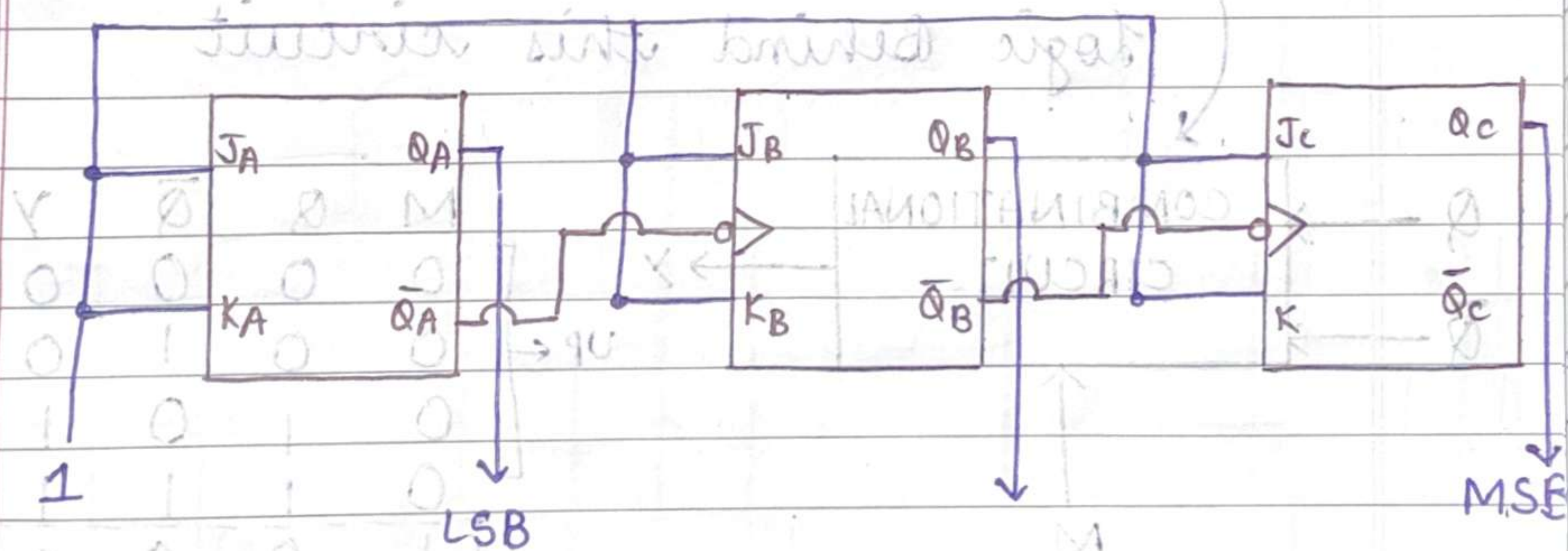
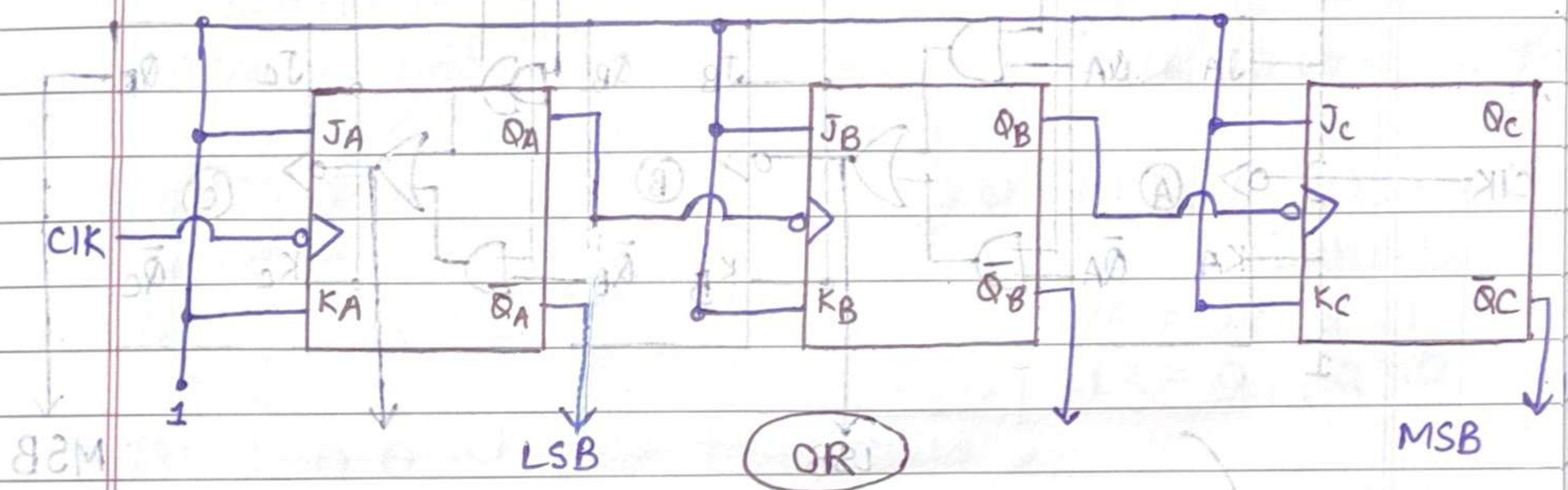
→ 4 Bit Asynchronous up counter



CLK	Q _D	Q _C	Q _B	Q _A	Binary
0	0	0	0	0	0
1	0	0	0	1	1
2	0	0	1	0	2
3	0	0	1	1	3
4	0	1	0	0	4
5	0	1	0	1	5
6	0	1	1	0	6
7	0	1	1	1	7
8	1	0	0	0	8
9	1	0	0	1	9
10	1	0	1	0	10
11	1	0	1	1	11

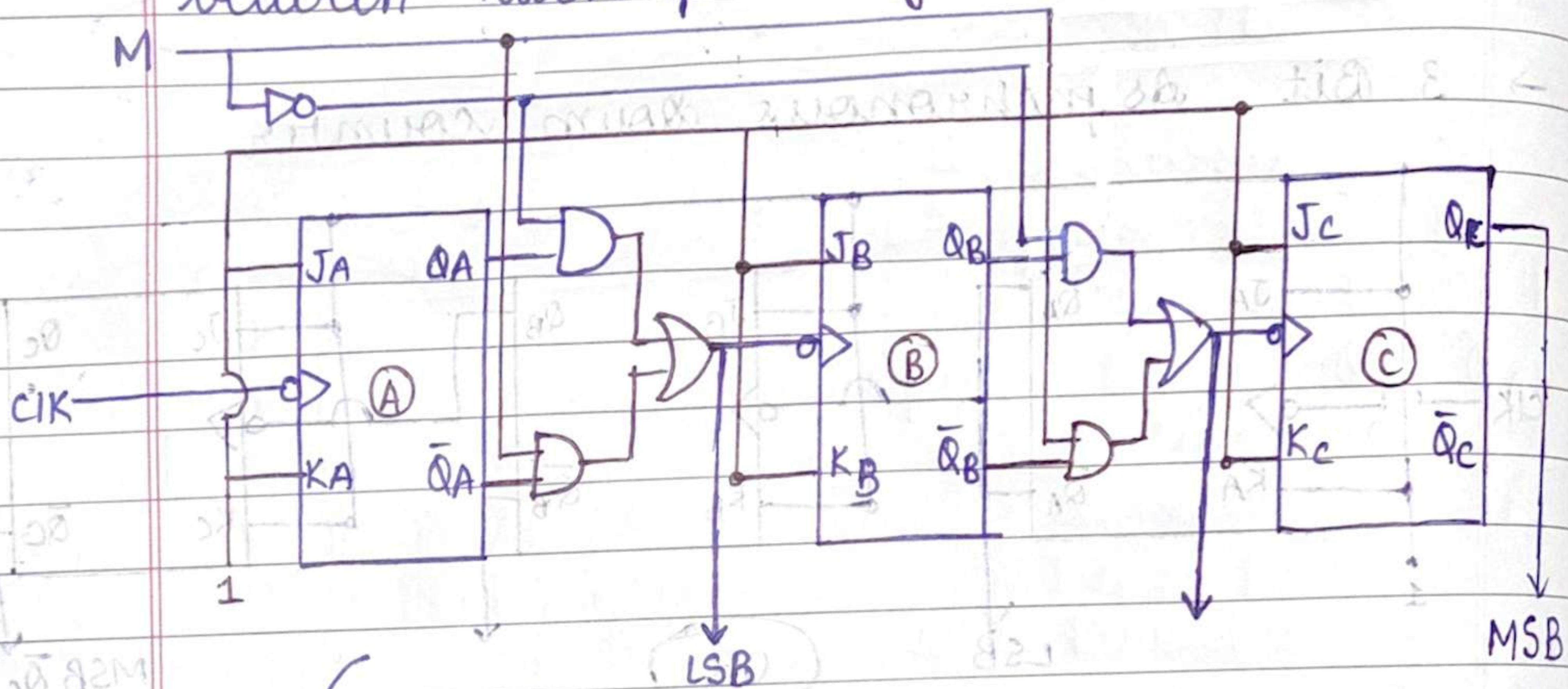
12 11 10 9 8 7 6 5 4 3 2 1
 13 12 11 10 9 8 7 6 5 4 3 2 1
 14 13 12 11 10 9 8 7 6 5 4 3 2 1
 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1
 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1

→ 3 Bit Asynchronous Down Counter

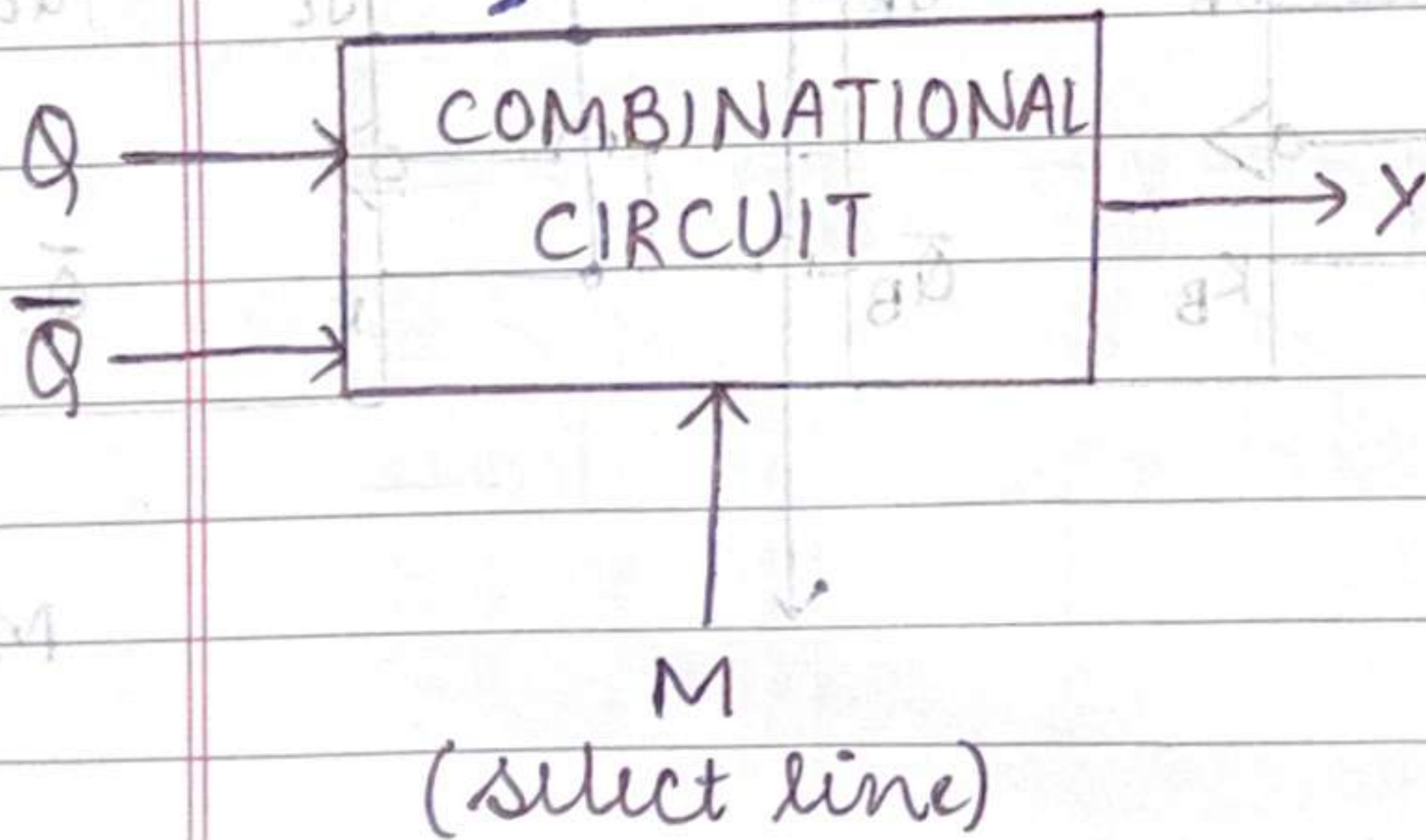


CLK	$\bar{Q}_C$	Q_B	$\bar{Q}_A$	Binary
0	1	1	1	7
1	1	1	0	6
2	1	0	1	5
3	1	0	0	4
4	0	1	1	3
5	0	1	0	2
6	0	0	1	1
7	0	0	0	0

- 3 Bit Asynchronous UP/DOWN counter
- A mode control input (M) is used to select either UP/DOWN mode
 - A combinational circuit is required between each pair of FF.



Logic Behind this circuit



M	Q	$\bar{Q}$	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

Q to CLK $\leftarrow M = 0 \rightarrow$ UP COUNTING

$M = 1 \rightarrow$ DOWN COUNTING

$\bar{Q}$ to CLK

M	Q	$\bar{Q}$	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

$$Y = \bar{M}Q + M\bar{Q}$$

→ Modulus of counter & MOD-n counters
n bit ripple counter is called MOD- 2^n counter (ASYNCHRONOUS)

2 bit ripple counter → MOD-4 counter

3 bit ripple counter → MOD-8 counter

→ MOD-6 counter using MOD-8 counter

status → 6 0101

max count → 6-1 = 5

status → 8

max count → 7

Use preset & clear inputs

PR = 0, Q = 1

CLR = 0, Q = 0

(0) 0 0 0

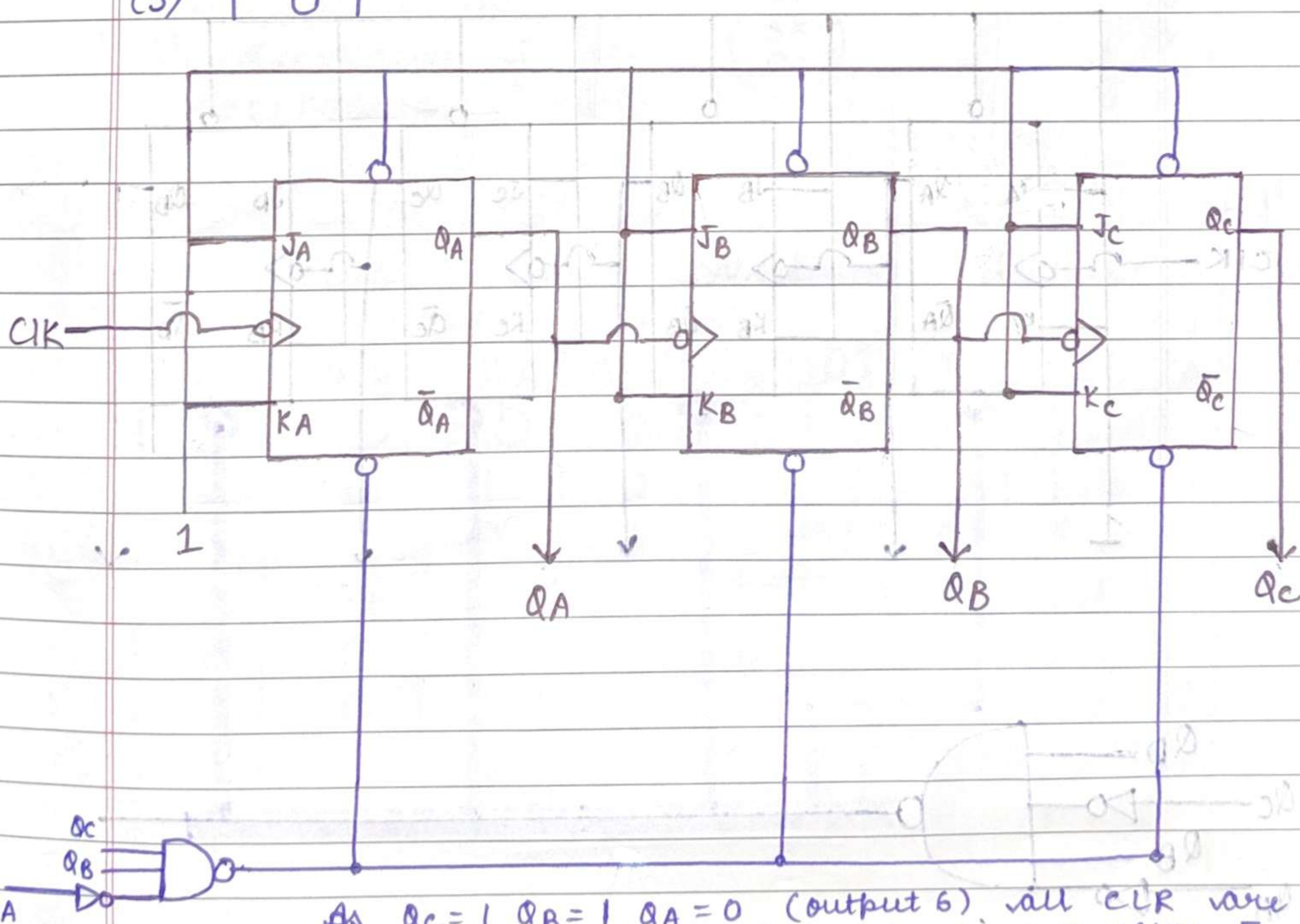
(1) 0 0 1

(2) 0 1 0

(3) 0 1 1

(4) 1 0 0

(5) 1 0 1



As $Q_C = 1$, $Q_B = 1$, $Q_A = 0$ (output 6) all CLR are set to 0 by NAND gate & counter gives 0 after 5

→ Decade / BCD Ripple counter

No of states = 10

Max count = $10 - 1 = 9$

0 0 0 0

0 0 0 1

0 0 1 0

0 0 1 1

0 1 0 0

0 1 0 1

0 1 1 0

0 1 1 1

1 0 0 0

1 0 0 1

1 0 1 0

1 0 1 1

We need to reset when we get a 10

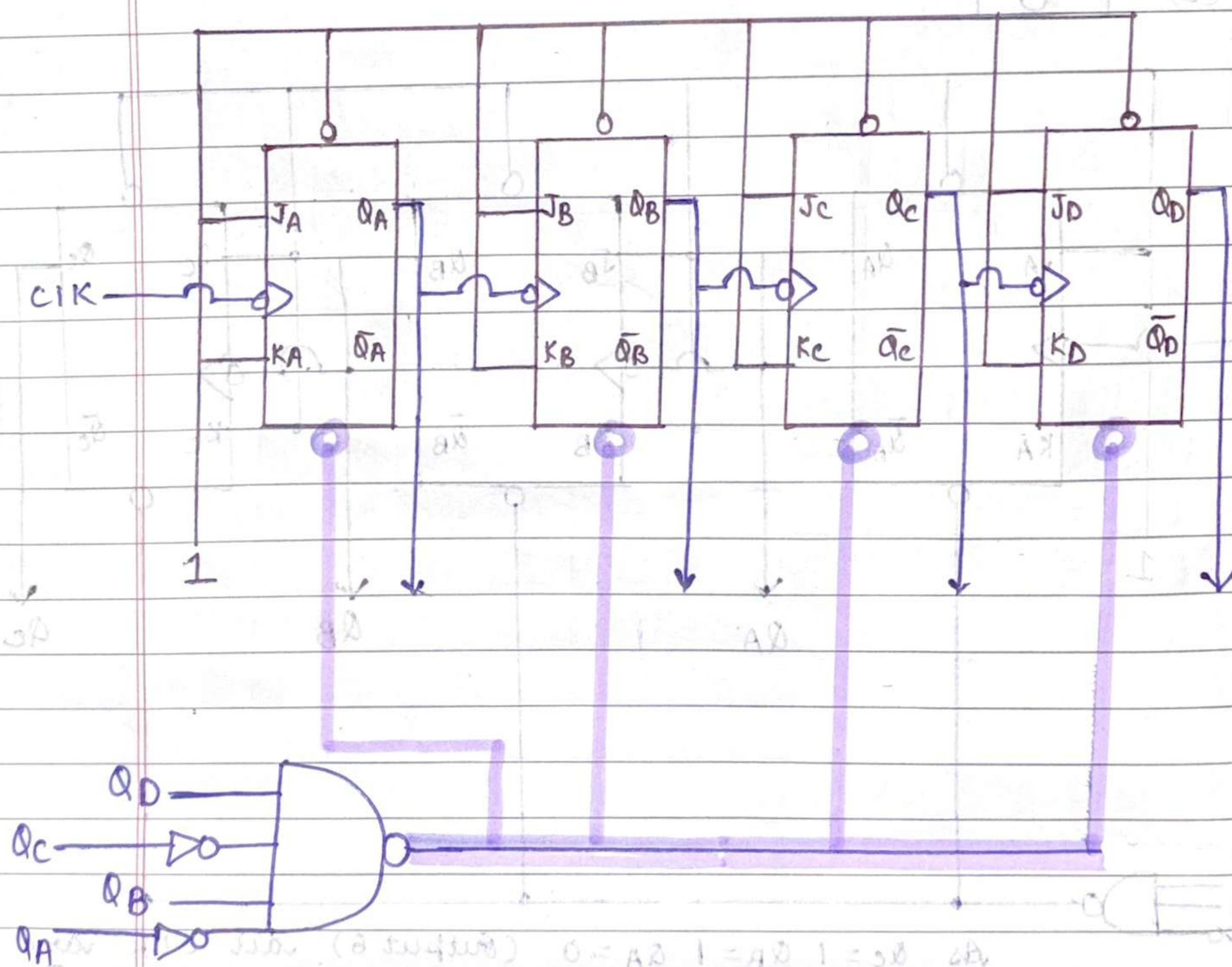
so

1010 (10)

$Q_D Q_C Q_B Q_A$

Use a NAND gate with inputs $Q_D, \bar{Q}_C, Q_B, \bar{Q}_A$ to set clear = 0

↓ useless
Reset the FF when 10 reaches



→ Synchronous counters

1. Decide the no of flip flops
2. Make excitation table of flip flop
3. State diagram & circuit table
4. Expression from K map
5. Draw logic Diagram

→ 3 bit synchronous UP counter

Q_n Q_{n+1} T

0 0 0

0 1 1

1 0 1

1 1 0

1 0 1

0 1 1

1 1 0

0 0 0

1 1 1

0 0 0

1 1 1

0 0 0

1 1 1

0 0 0

1 1 1

0 0 0

1 1 1

0 0 0

1 1 1

0 0 0

1 1 1

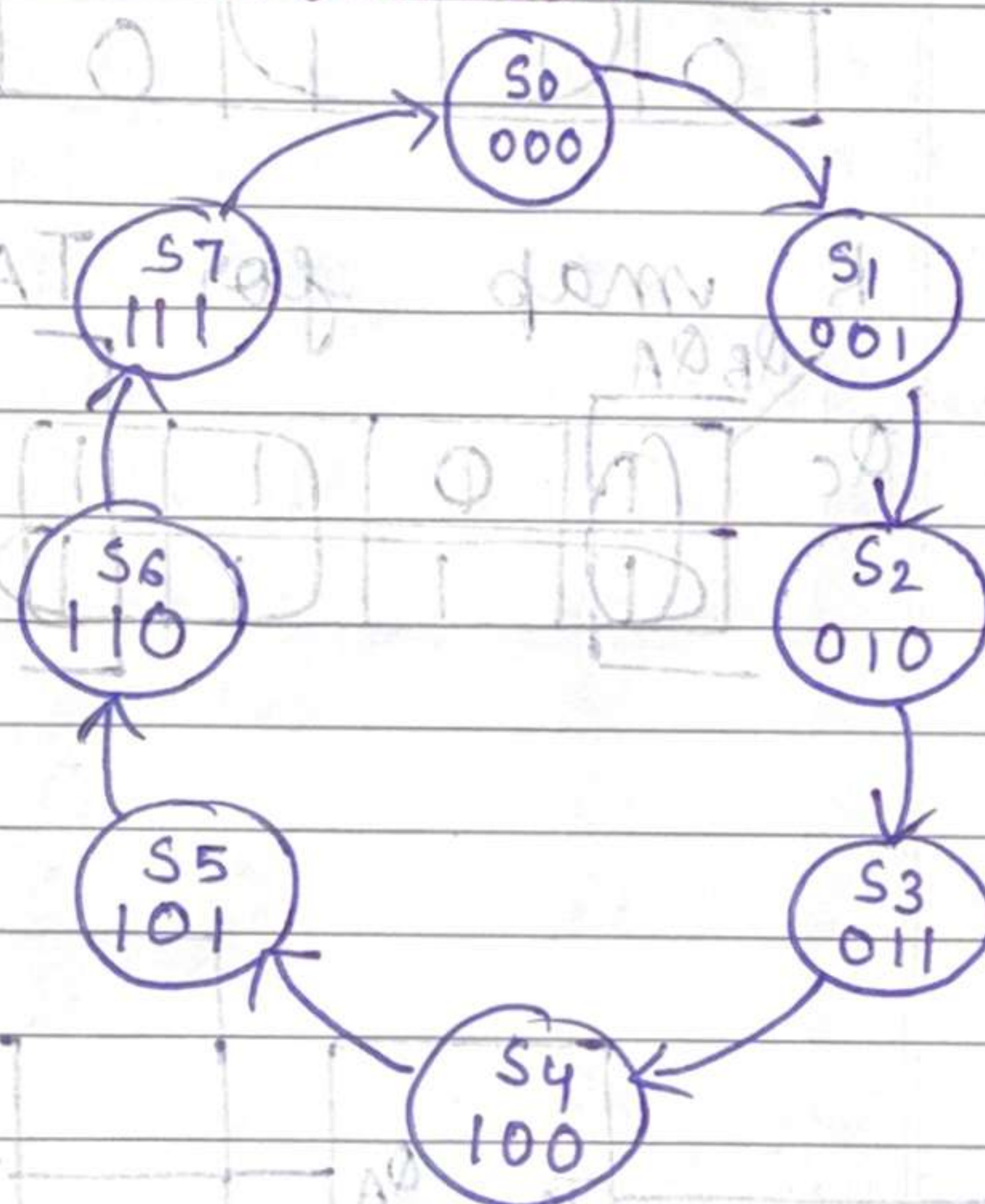
0 0 0

1 1 1

0 0 0

1 1 1

To find T_c
see Q_c & Q_c^+
& compare from
excitation table



Present State			Next State			Values		
Q_c	Q_B	Q_A	Q_c^+	Q_B^+	Q_A^+	T_c	T_B	T_A
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	1
0	1	0	0	1	1	0	0	1
0	1	1	1	0	0	1	1	1
1	0	0	1	0	1	0	0	1
1	0	1	1	1	0	0	1	1
1	1	0	1	1	1	0	0	1
1	1	1	0	0	0	1	1	1

K map for T_C

$Q_C \backslash Q_B Q_A$	00	01	11	10
0	0	0	1	0
1	0	0	1	0

$$T_C = Q_B Q_A$$

K map for T_B

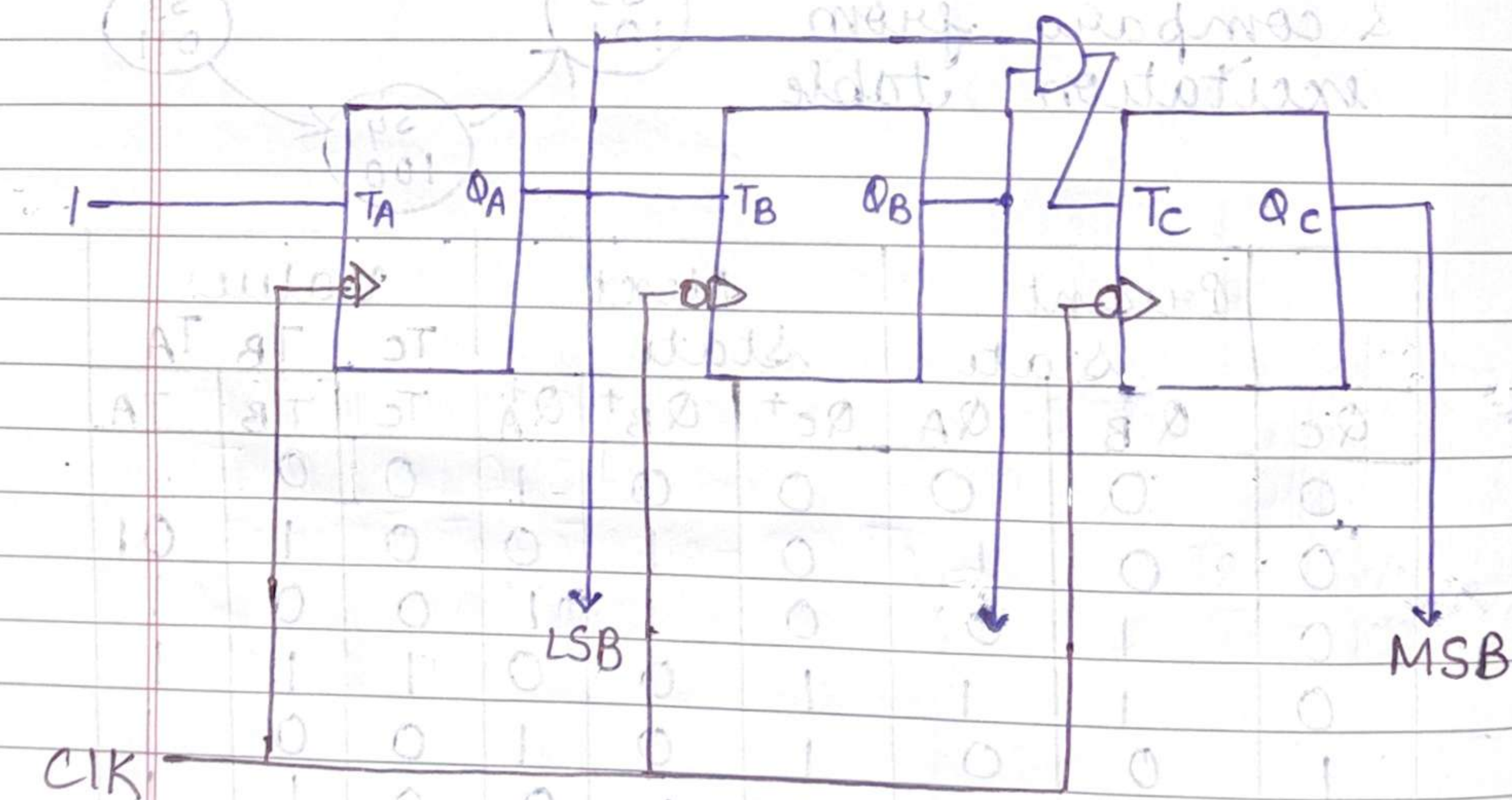
$Q_C \backslash Q_B Q_A$	00	01	11	10
0	0	1	1	0
1	0	1	1	0

$$T_B = Q_A$$

K map for T_A

$Q_C \backslash Q_B Q_A$	00	01	11	10
0	1	1	1	1
1	1	1	1	1

$$T_A = 1$$



UP COUNT $\rightarrow 0 = M$
DOWN COUNT $\rightarrow 1 = M$

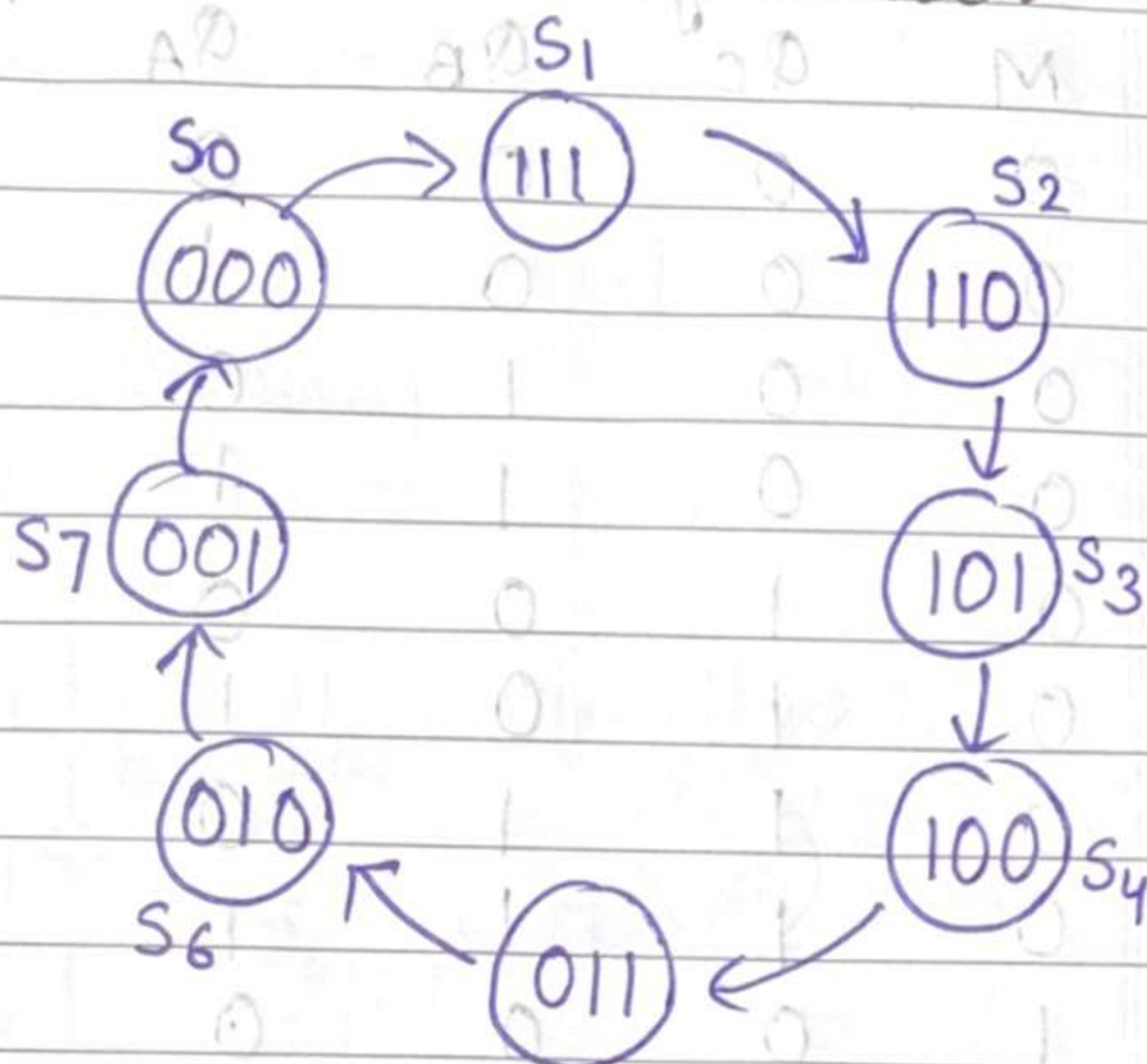
classmate

Date

Page

→ 3 bit synchronous down counter

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0



Present			Next			Values		
Q_C	Q_B	Q_A	Q_C^+	Q_B^+	Q_A^+	T_C	T_B	T_A
0	0	0	1	1	1	0	1	1
1	1	1	1	0	0	0	0	1
1	0	1	1	0	0	0	1	1
1	0	0	1	0	0	0	0	1
1	0	0	0	1	1	1	1	1
0	1	1	0	1	0	0	0	1
0	1	0	0	0	1	0	0	1
0	0	1	0	0	0	0	0	1

Q_C	Q_A	Q_B	Q_C	Q_A	Q_B	Q_C	Q_A	Q_B
1	0	0	1	0	0	1	1	1
1	0	0	1	0	0	1	1	1

For T_C

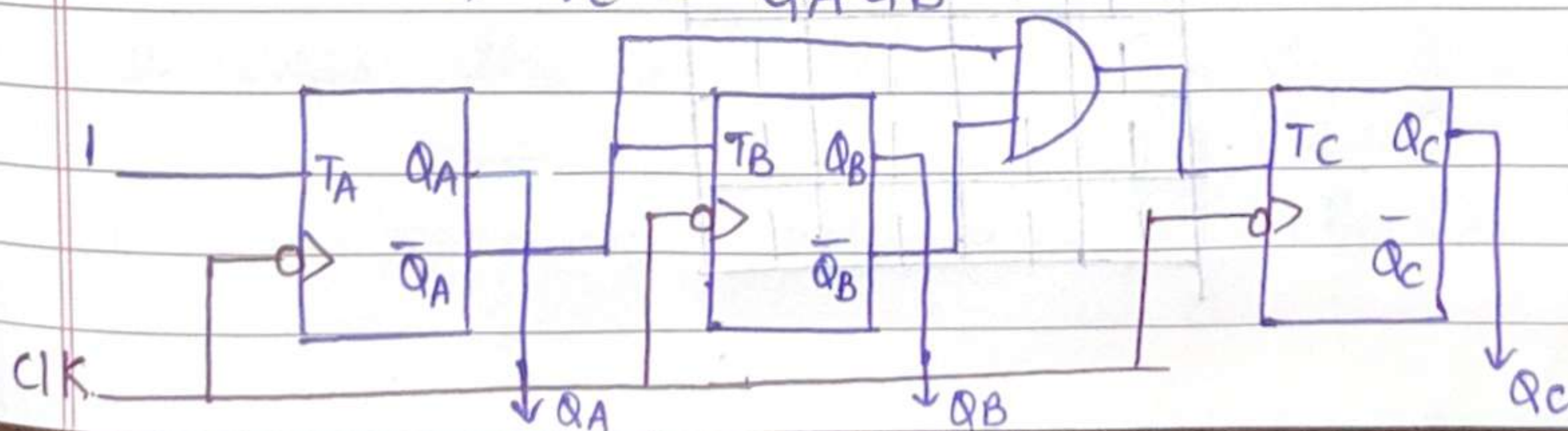
For T_B

For T_A

$$T_A = 1$$

$$\Rightarrow T_B = \bar{Q}_A$$

$$\Rightarrow T_C = \bar{Q}_A \bar{Q}_B$$



$M=0 \rightarrow$ UP COUNT
 $M=1 \rightarrow$ DOWN COUNT

$\rightarrow$ 3 bit synchronous up/down counter

M	Q_C	Q_B	Q_A	Q_C^+	Q_B^+	Q_A^+	T_C	T_B	T_A
0	0	0	0	0	0	1	0	0	1
0	0	1	0	0	1	0	0	1	1
0	0	1	1	0	1	1	0	1	1
0	0	0	1	1	0	0	1	1	1
0	1	0	0	1	0	1	0	0	1
0	1	0	1	1	1	0	0	1	1
0	1	1	0	1	1	1	0	0	1
0	1	1	1	0	0	0	1	1	1
1	0	0	0	1	1	1	1	1	1
1	0	0	1	0	0	0	0	0	1
1	0	1	0	0	1	0	0	1	1
1	0	1	1	0	1	1	0	1	1
1	1	0	0	1	0	0	1	1	1
1	1	0	1	1	0	1	1	1	1
1	1	1	0	1	1	1	1	1	1
1	1	1	1	0	0	0	0	0	1

$M \backslash Q_B Q_A$	$\bar{Q}_B \bar{Q}_A$	$\bar{Q}_B Q_A$	$Q_B \bar{Q}_A$	$Q_B Q_A$
$\bar{M} \bar{Q}_C$	0	0	1	0
$\bar{M} Q_C$	0	0	1	0
$M \bar{Q}_C$	1	0	0	0
$M Q_C$	1	0	0	0

0	1	1	0
0	1	1	0
1	0	0	1
1	0	0	1

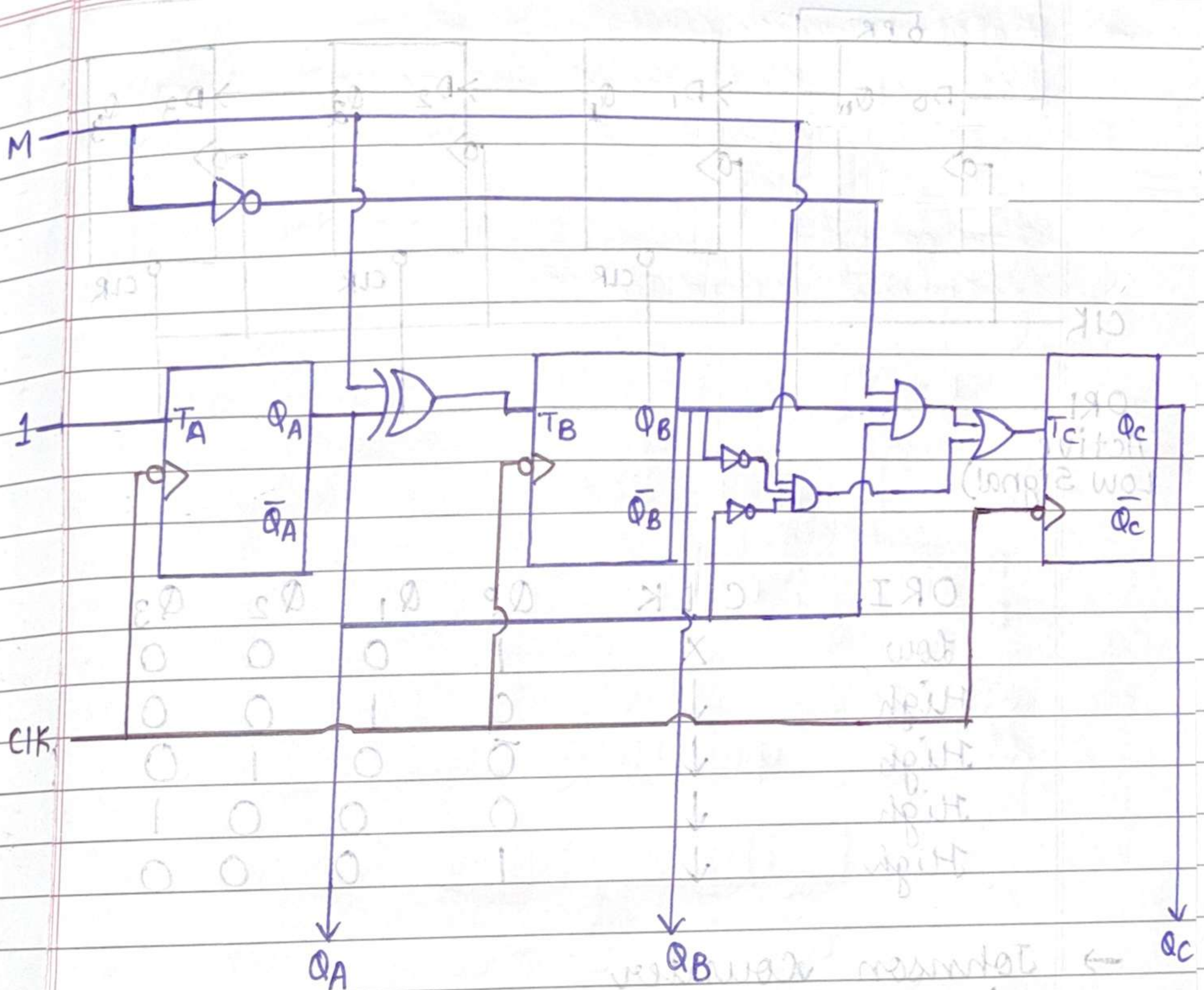
$T_C = M \bar{Q}_B \bar{Q}_A + \bar{M} Q_B Q_A$

$T_B = \bar{M} Q_A + M \bar{Q}_A$

$T_B = M \oplus Q_A$

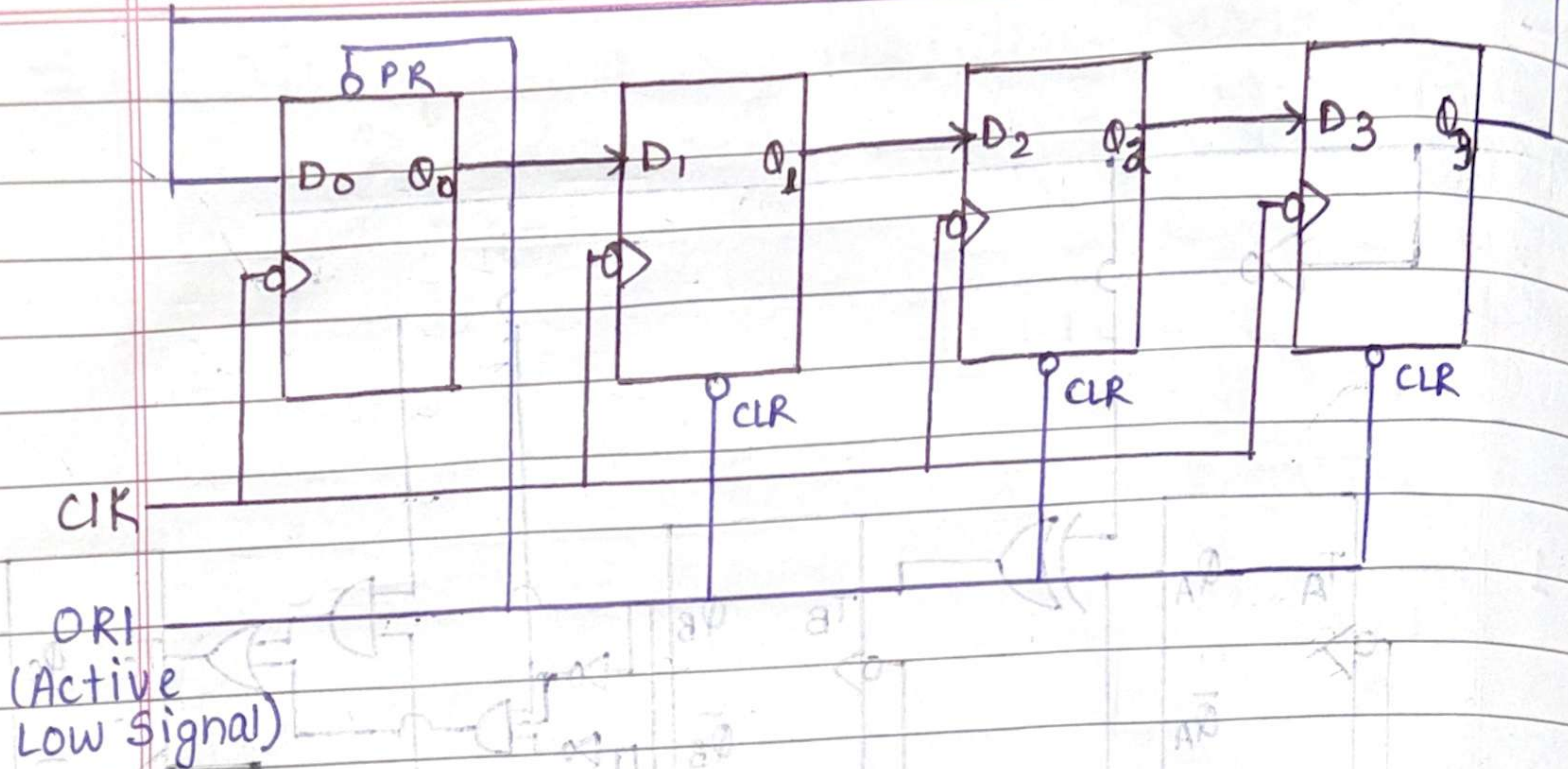
$T_A = 1$

1	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1



→ Ring Counter

- Ring counter is a typical explanation of shift registers
- The only change is that the out of last flip flop is connected to the input of first flip flop
- No of states = no of flip flops used
- It uses the overriding inputs (ORI)
 - PRESET $PR=0$ $Q=1$
 - CLEAR $CLR=0$ $Q=0$



ORI	CLK	Q ₀	Q ₁	Q ₂	Q ₃
Low	X	1	0	0	0
High	↓	0	1	0	0
High	↓	0	0	1	0
High	↓	0	0	0	1
High	↓	1	0	0	0

→ Johnson Counter

↳ TWISTED TAIL RING COUNTER
 ↳ SWITCH TAIL RING COUNTER

} ALSO called
 Johnson
 counter

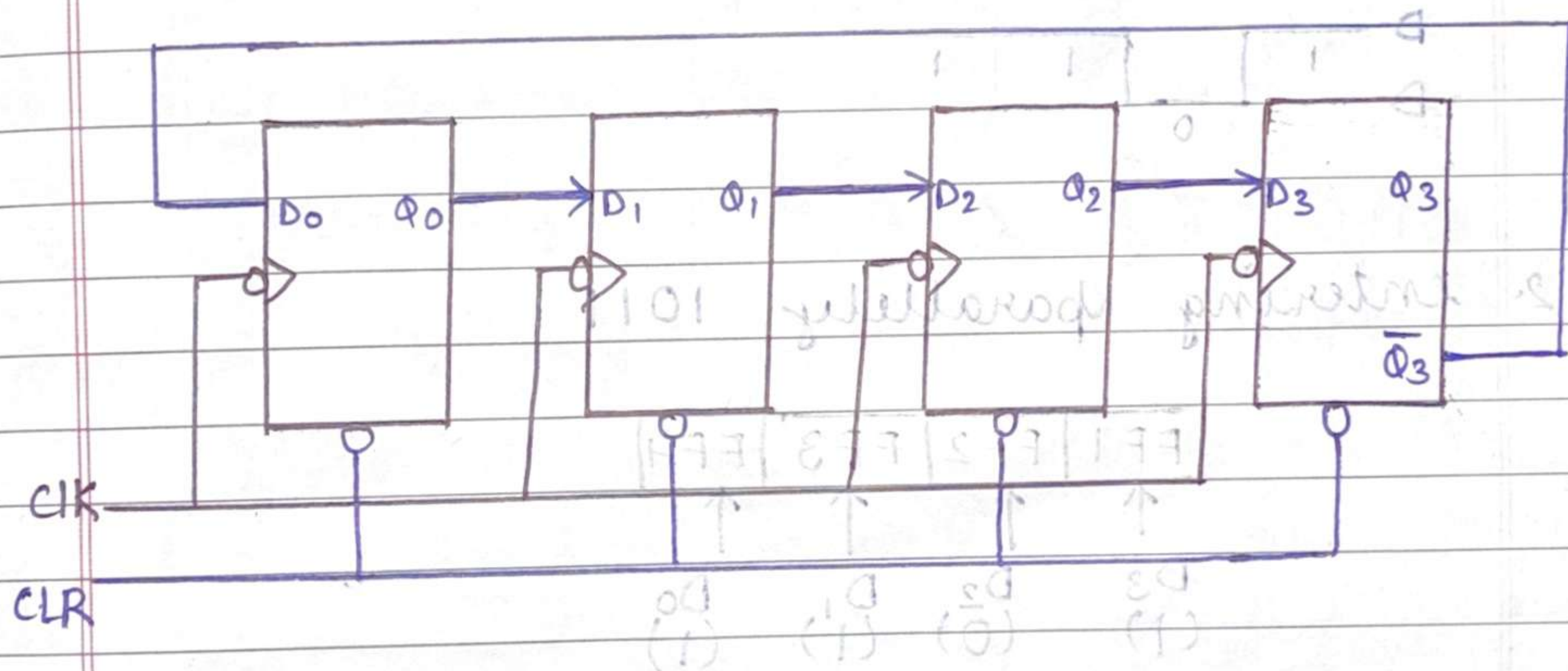
- No of states = $2 \times$ no of flip flops
- Uses only clear input for all the flip flops
- Q_3 is used as input to first flip flop

(ORI) is used as input to first flip flop

PR = 0
 CLR = 0

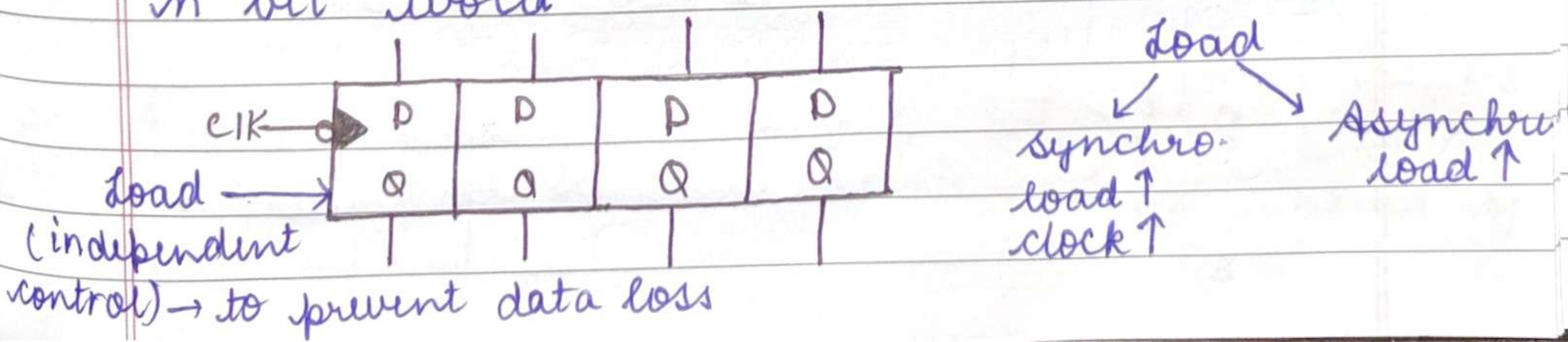
CLEAR
 T₁ = 0

CLR	CLK	Q ₀	Q ₁	Q ₂	Q ₃
LOW	X	0	0	0	0
↓	↓	1	0	0	0
↓	↓	1	1	0	0
↓	↓	1	1	1	0
↓	↓	0	1	1	1
↓	↓	0	0	1	1
↓	↓	0	0	0	1
↓	↓	0	0	0	0



→ Registers

- Flip flop is a one bit memory cell.
- To ↑ the storage capacity, we have to use group of flip flops. This group of flip flops is known as register.
- The n bit register consists of n number of flip flops and is capable of storing n bit word.



→ Data formats

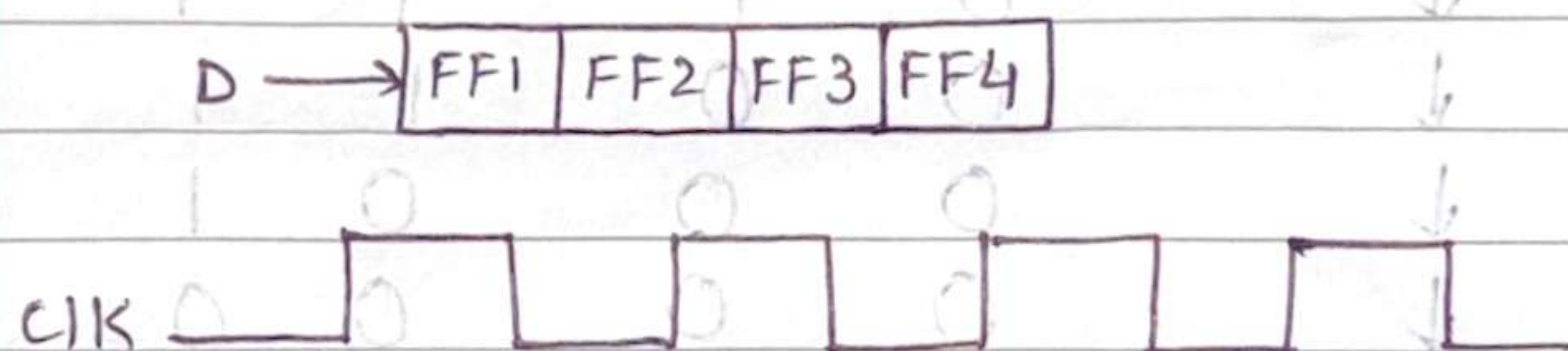
Data can be entered in serial / parallel for

↓
one bit at a time
(TEMPORAL CODE)

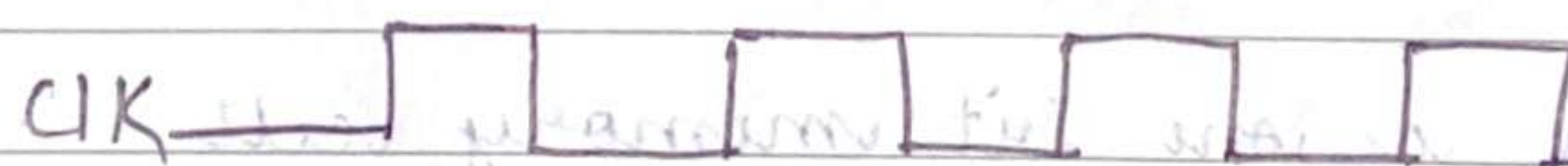
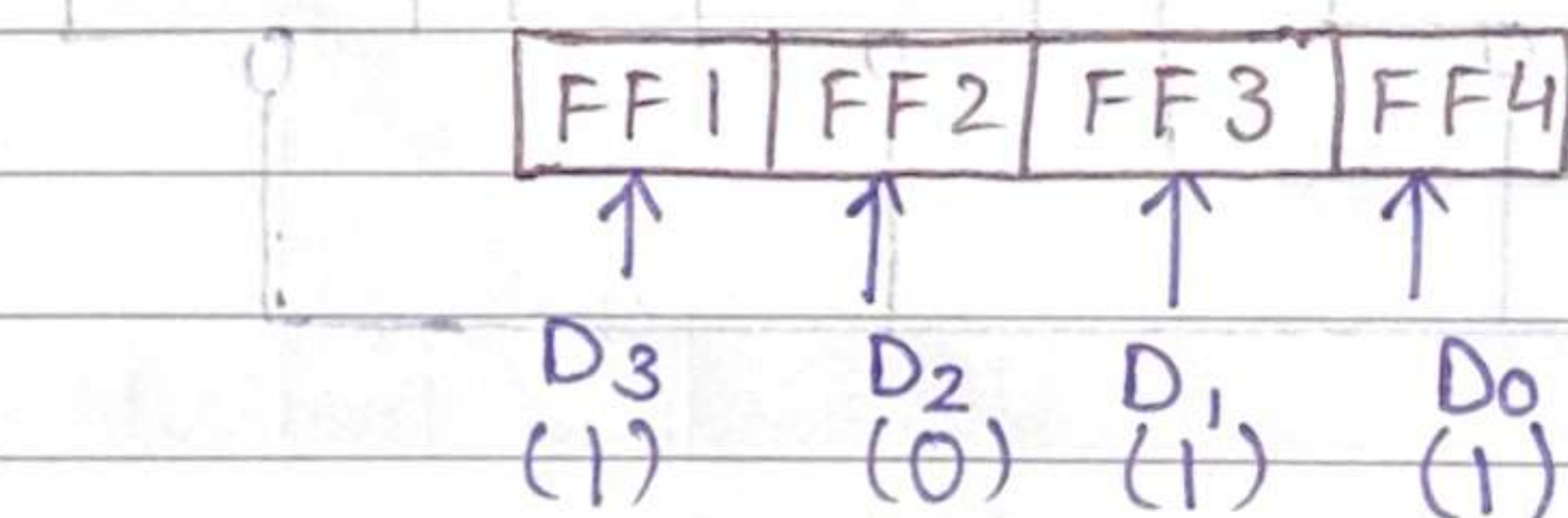
↳ all bits at a time
(SPECIAL CODE)

For eg :-

1. entering serially 1011



2. entering parallelly 1011



D3 (1)

D2 (0)

D1 (1)

D0 (1)

D3 (1)

D2 (0)

D1 (1)

D0 (1)

D3 (1)

D2 (0)

D1 (1)

D0 (1)

→ Classification of registers

1. depending on input/output

→ SISO serial input serial output

→ SIPO serial input parallel output

→ PISO parallel input serial output

→ PIPO parallel input parallel output

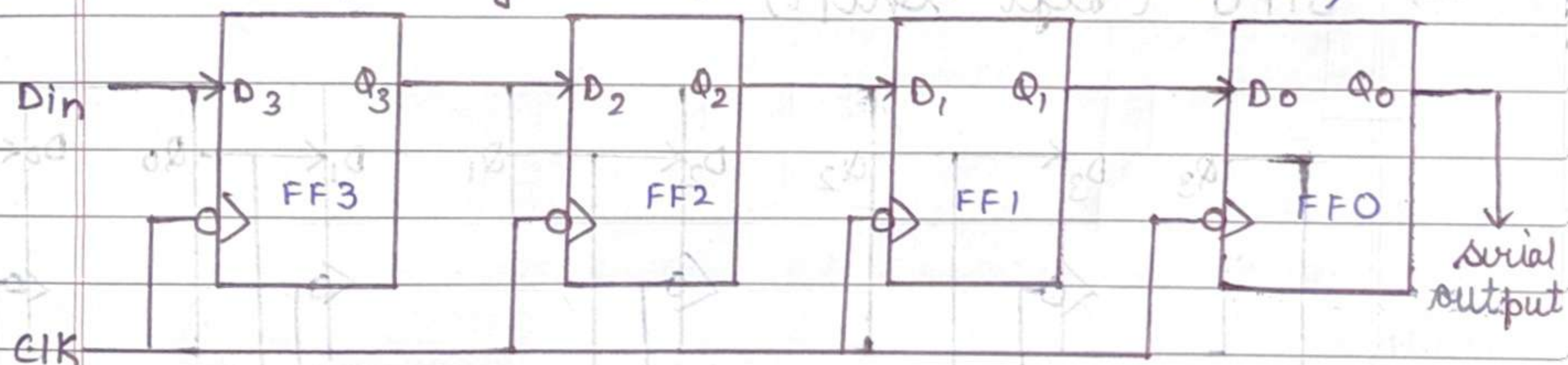
2. depending on applications

→ Shift Register

→ Storage Register (PIPO) → Buffer Register

→ Shift Registers (SISO) → shift right
→ shift left

Shift Right (input at D_3 , output at D_0)



CLK	Q_3	Q_2	Q_1	Q_0
Initially	0	0	0	0
↓	1	0	0	0
↓	1	1	0	0
↓	1	1	1	0
↓	1	1	1	1

To Store 1111

Give LSB to

D_3

⇒ $D_3 = 1$

& rest 0

then next bit upto

MSB



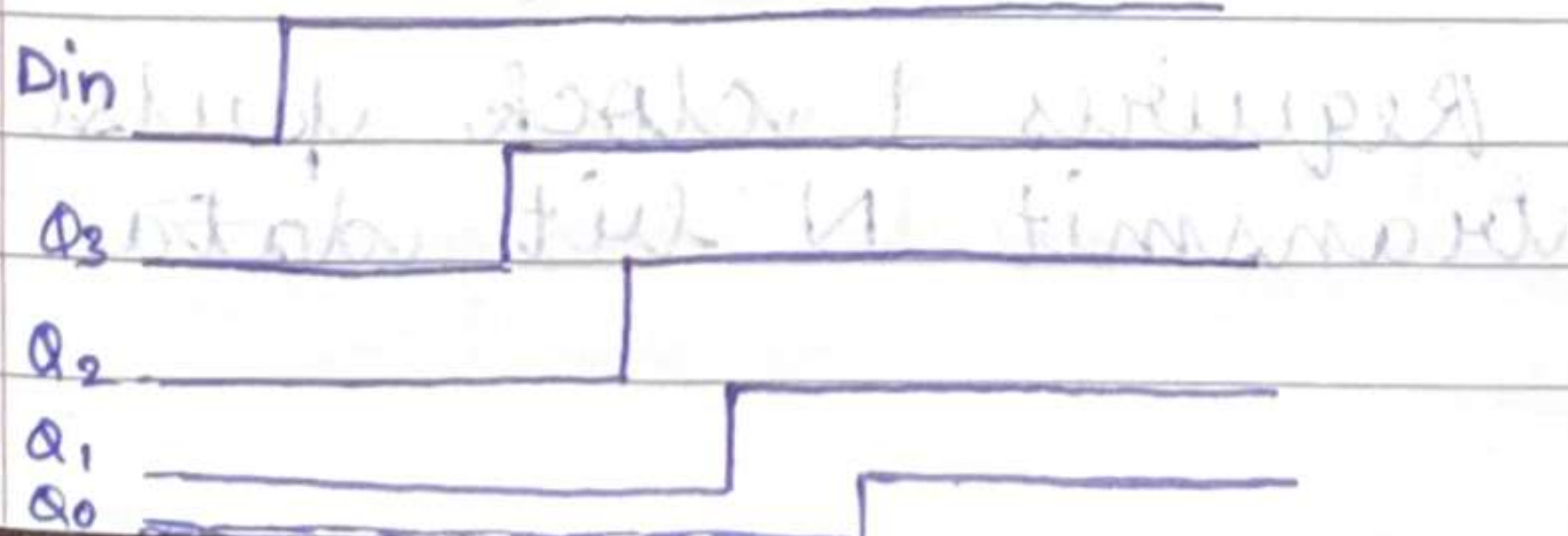
For shift left

INPUT AT D_0 &

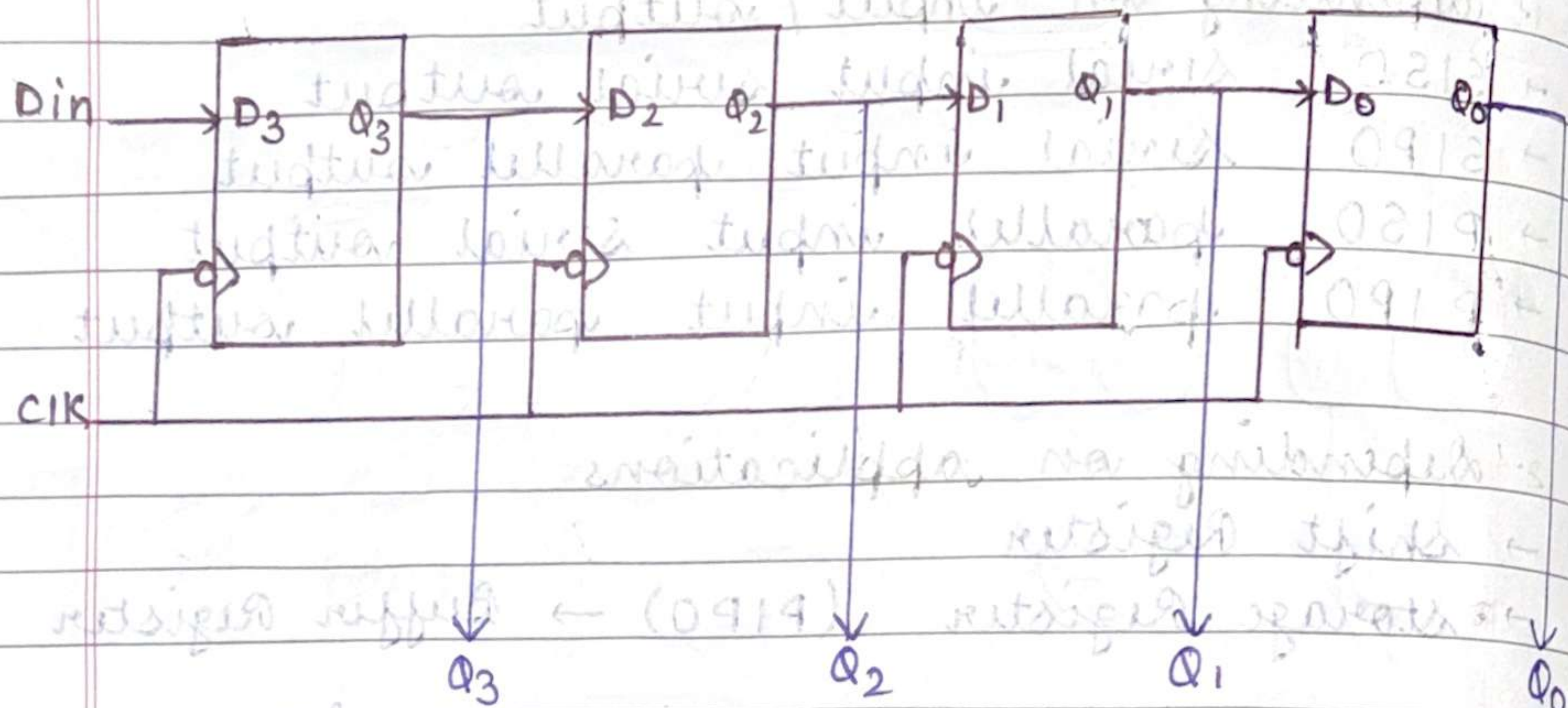
OUTPUT at D_3

give LSB to

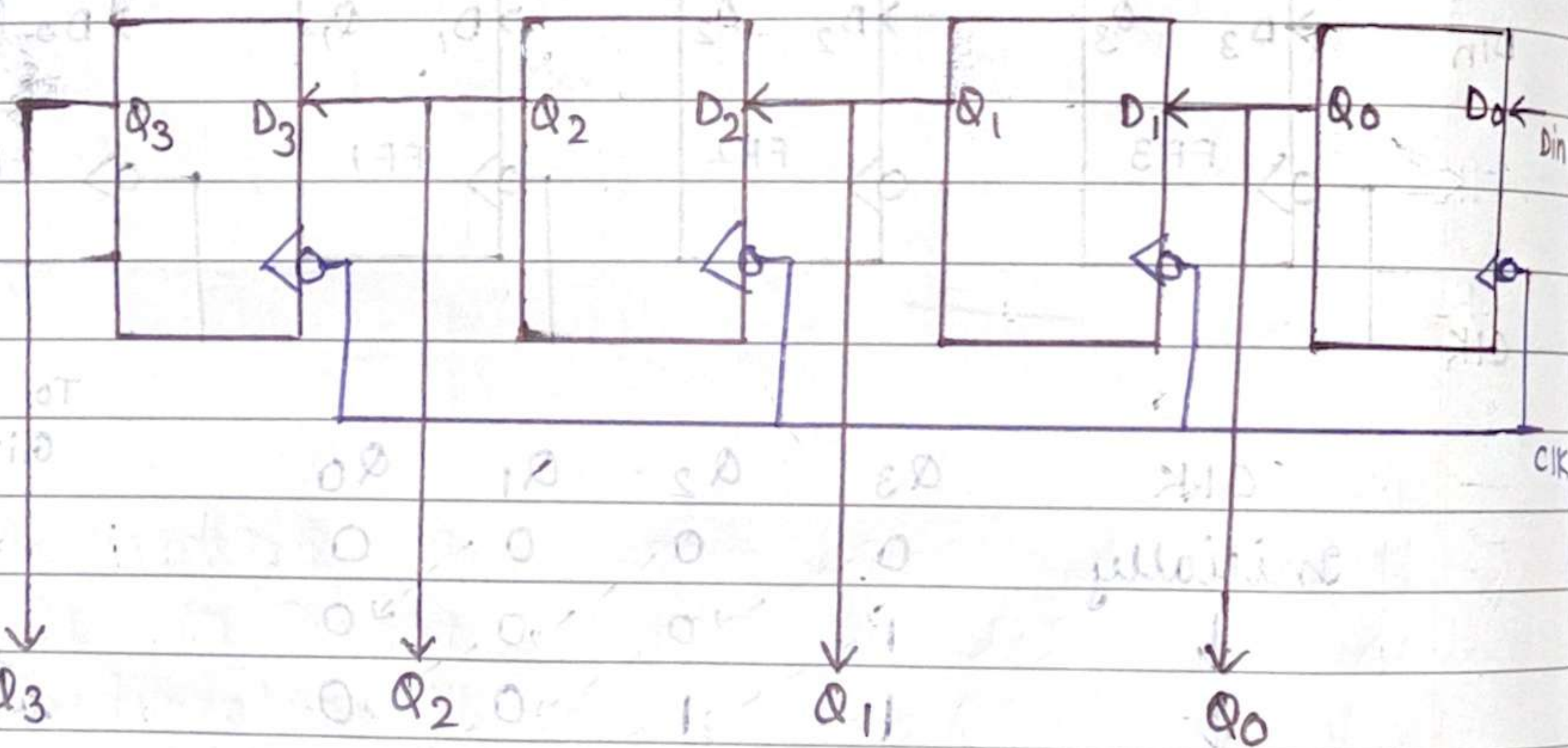
D_0 instead of D_3



→ SIPO (Right shift)



→ SIPO (Left shift)



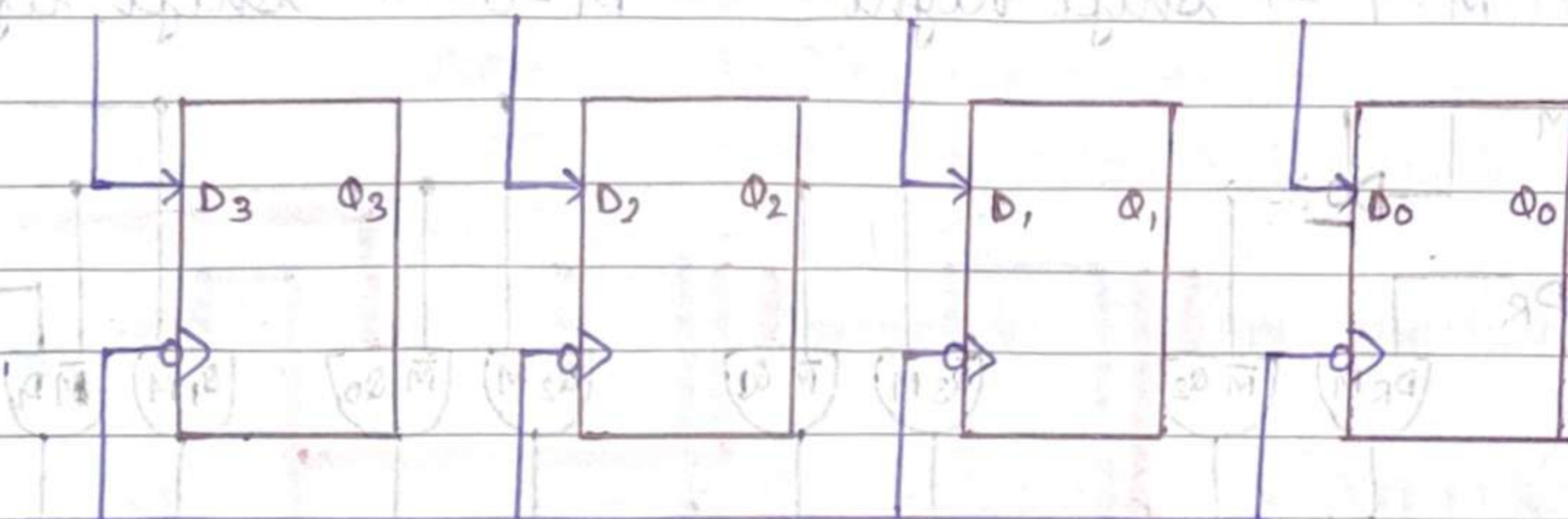
SISO
SIPO

Requires N clock pulses
in order to transmit
N bit data

PIPO →

Requires 1 clock pulse to
transmit N bit data

→ PIPO (Shift Right)



→ PISO

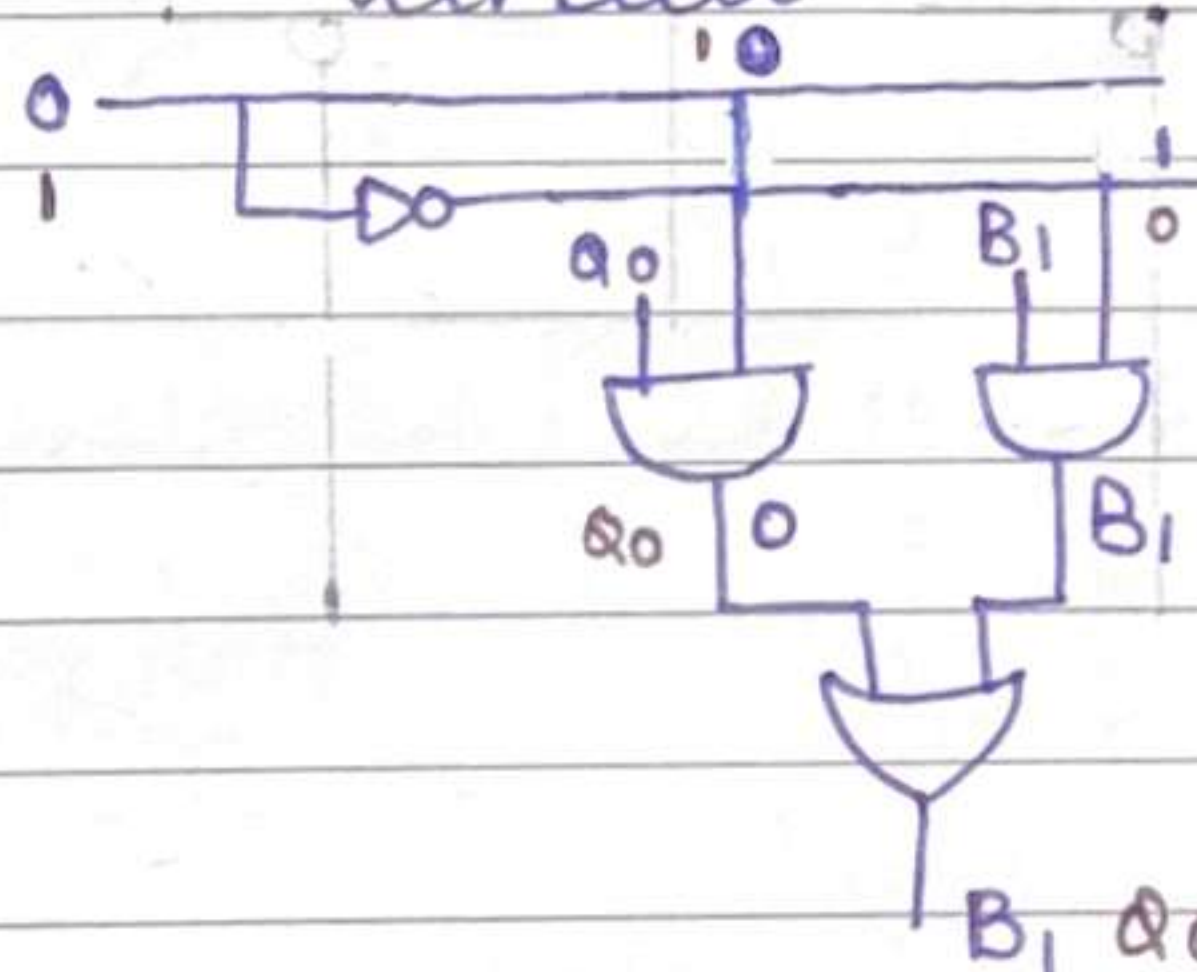
It has 2 modes i.e. load mode & shift mode.

(M=0)

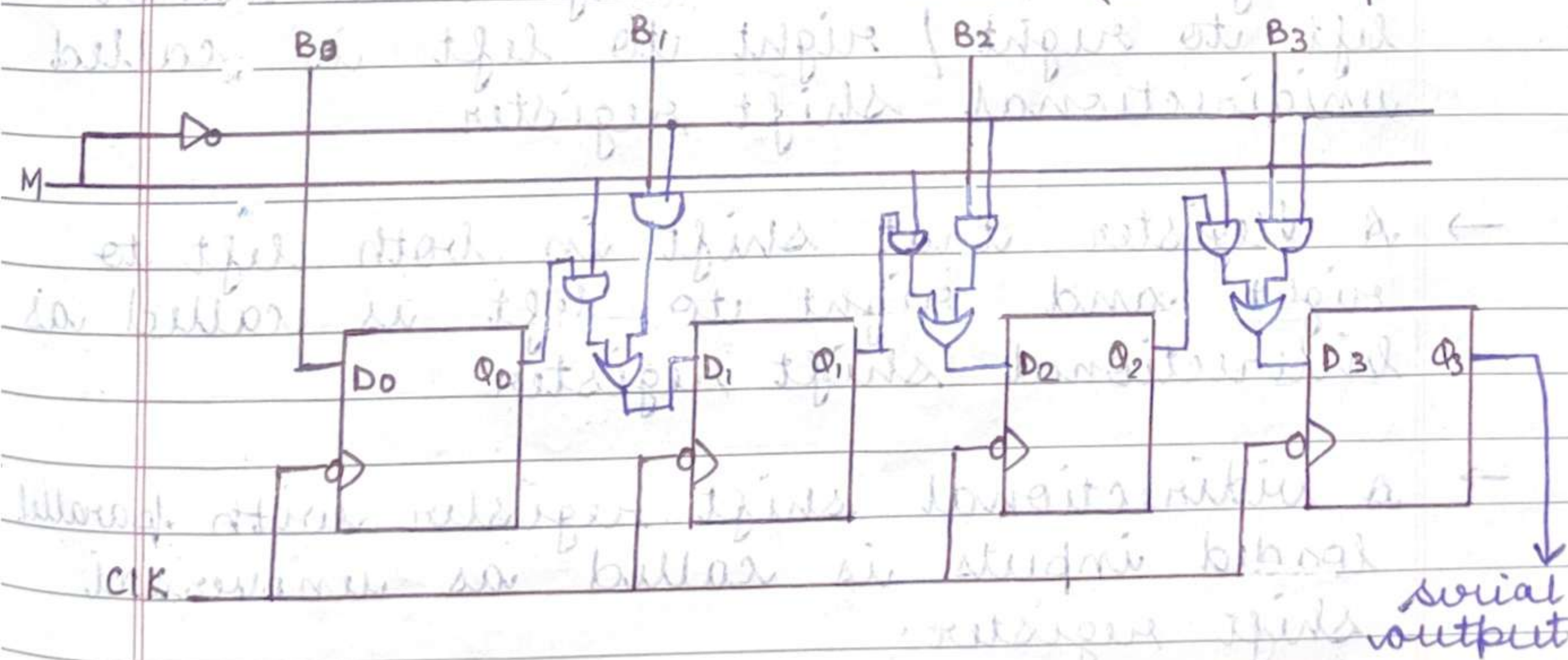
(M=1)

used to store data simultaneously using combinational circuit

to give serial output

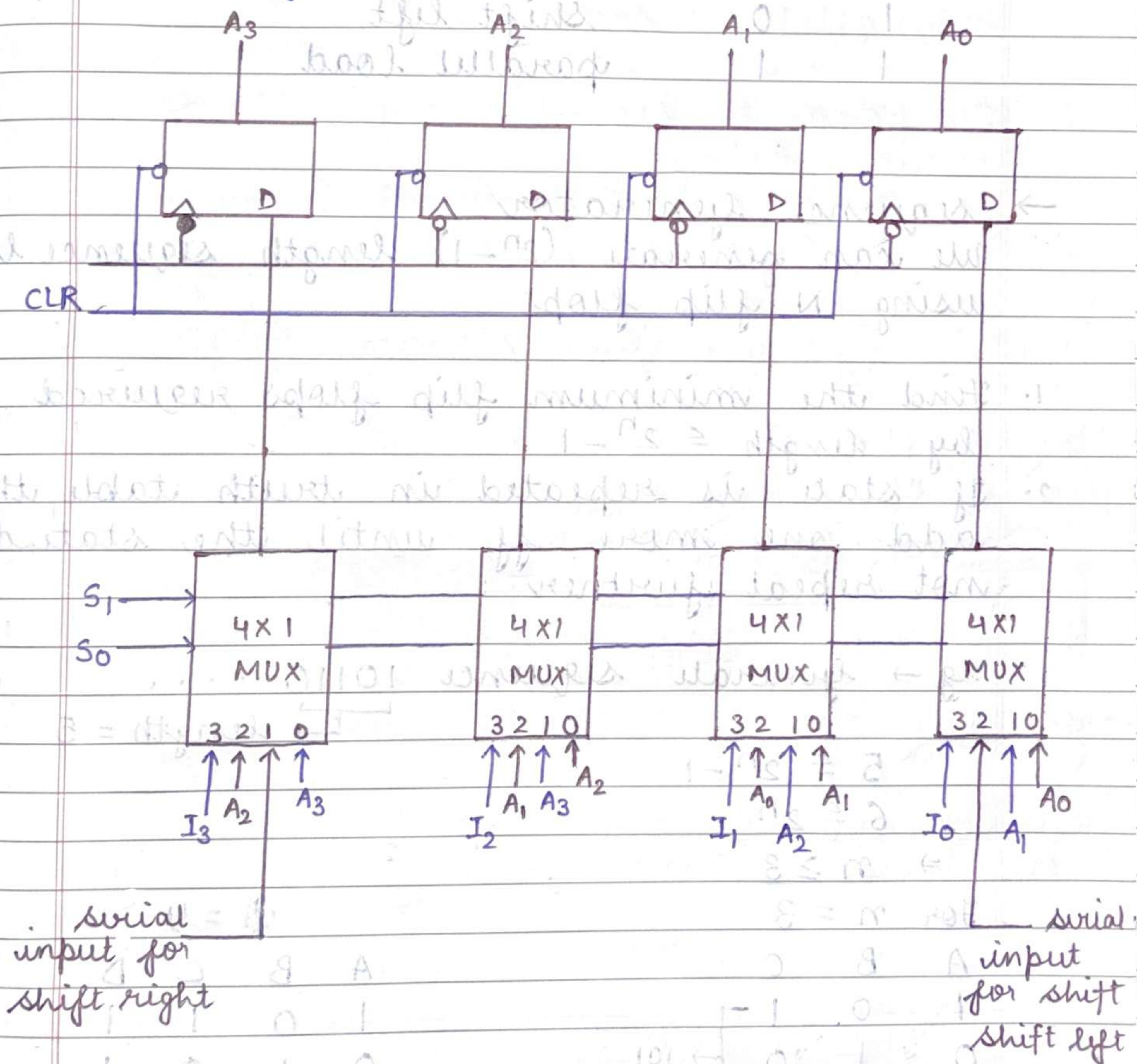


whatever is Q0 every time B1 is loaded shift mode



→ Universal Shift Register

It's capable to perform SISO, SIPO, PISO, PIPO all using one circuit



0th index of MUX $\rightarrow A_n$ - ($n \rightarrow$ no. of bit)

1st index of MUX $\rightarrow A_{n+1}$ / serial input (right)

2nd index of MUX $\rightarrow A_{n-1}$ / serial input (left)

3rd index of MUX $\rightarrow I_n$ (input)

S_1	S_0	Operation
0	0	No change
0	1	Shift right
1	0	Shift left
1	1	parallel load

→ Sequence generator

We can generate $(2^n - 1)$ length sequence by using N flip flops

1. Find the minimum flip flops required by length $\leq 2^n - 1$
2. If state is repeated in truth table then add one more ff until the state does not repeat further

eg → generate sequence 10110...
length = 5

$$5 \leq 2^n - 1$$

$$6 \leq 2^n$$

$$\Rightarrow n \geq 3$$

For $n = 3$

A B C

1 0 1

0 1 0

1 0 1

0 1 0

1 0 1

0 1 0

1 0 1

0 1 0

1 0 1

0 1 0

repeats

add 1

flip

flop

top of next

$n = 4$

A B C D

1 0 1 1

0 1 0 1

1 0 1 0

0 1 0 1

1 0 1 0

0 1 0 1

1 0 1 0

0 1 0 1

1 0 1 0

0 1 0 1

no repetition
 $\Rightarrow$ 4 flip flops will be used.

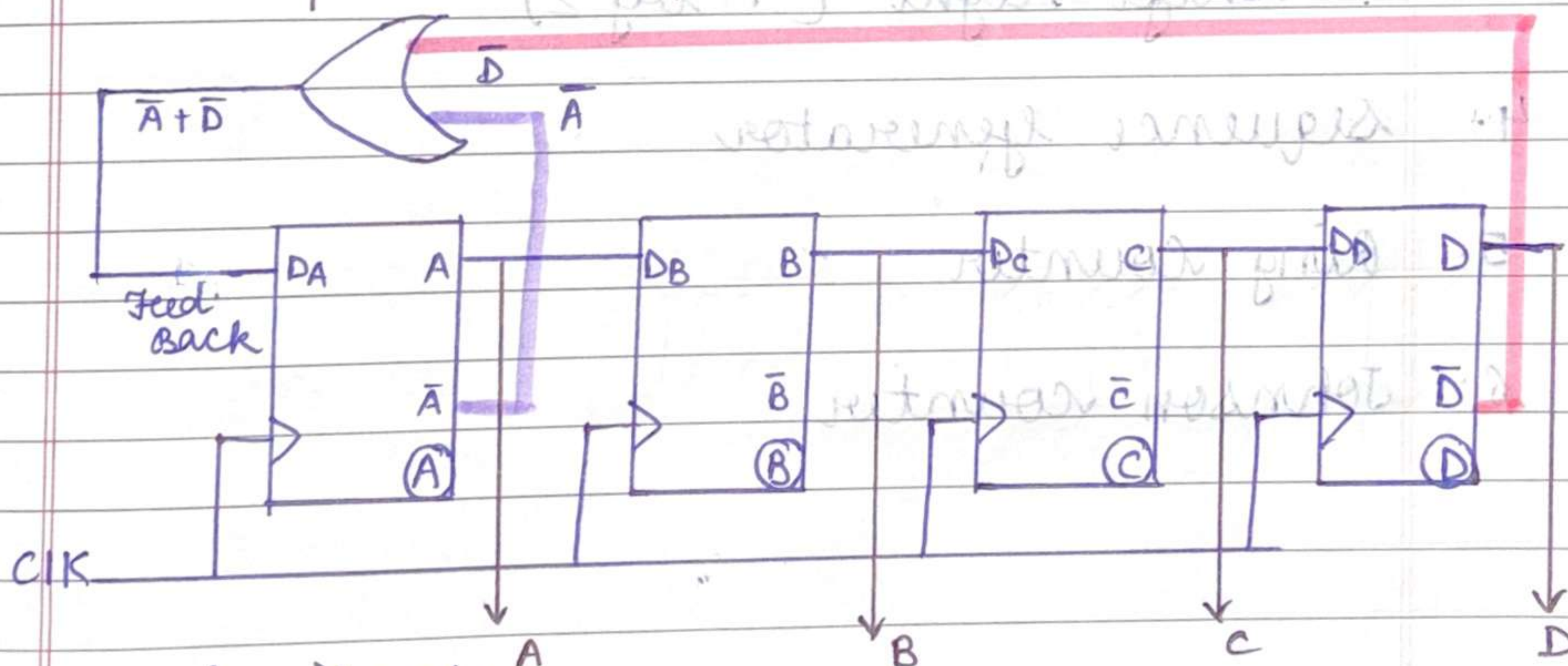
3. Now find $(F) \rightarrow$ similarly by shifting the last (D)

A	B	C	D	F (Feedback)
1	0	1	1	0 $\leftarrow A\bar{B}CD$
0	1	0	1	1 $\leftarrow \bar{A}B\bar{C}D$
1	0	1	0	1 $\leftarrow A\bar{B}C\bar{D}$
1	1	0	1	0 $\leftarrow AB\bar{C}D$
0	1	0	0	1 $\leftarrow \bar{A}BCD$

$\rightarrow$ Rest X

AB \ CD	$\bar{A}\bar{B}$	$\bar{A}B$	AB	$A\bar{B}$
$\bar{C}\bar{D}$	X	X	X	X
$\bar{C}D$	X	1	0	X
CD	X	1	X	0
$C\bar{D}$	X	X	X	1

$$F = \bar{C}\bar{D} + \bar{A}$$



similarly,

any sequence can be generated using these steps

$\rightarrow$ This is also an application of shift registers

→ Applications of Shift Registers

1. Data format conversion

→ Serial → Parallel

→ Parallel → Serial

2. Temporal & signal manipulation

→ Time delay

↳ SISO delays serial binary signal by N clock cycles

→ Temporary data storage / buffering

3. Arithmetic operations

→ Shift left ($\times$ by 2)

→ Shift right ($\div$ by 2)

4. Sequence generator

5. Ring Counter

6. Johnson counter

①

any register can be used as a shift register

for applications like data storage, data transfer, etc.